ANNA UNIVERSITY, CHENNAI

UNDER GRADUATE CURRICULUM (NON-AUTONOMOUS AFFILIATED INSTITUTIONS)

Programme: B.E., Mechanical Engineering

Regulations: 2025 (Revised 1, 2026), with effect from the AY 2026 – 27 to all the students of UG Programme.

Abbreviations:

HUM – Humanities (Languages, Management, Heritage, and others)

BS – Basic Science (Mathematics, Physics, Chemistry)

ES – Engineering Science (General (**G**), Programme Core (**PC**), Programme Elective (**PE**) & Emerging Technology (**ET**))

SD – Skill Development

SL – Self Learning

CDP–Capstone Design Project

OE – Open Elective

L –Laboratory Course

T – Theory

LIT –Laboratory Integrated Theory

PW – Project Work

IPW–Internship cum Project Work

DIC – Department Introductory Course

TCP –Total Contact Period(s)

Program Outcomes

1. **Engineering Knowledge**: Apply math, science, and engineering fundamentals to complex problems.
2. **Problem Analysis**: Identify and analyze complex problems using research and sustainability principles.
3. **Design Solutions**: Design systems and processes considering health, safety, cost, culture, and environment.
4. **Investigations**: Use experiments, modelling, and data analysis to reach valid conclusions.
5. **Engineering Tools**: Apply modern tools for modelling and problem-solving, recognizing their limits.
6. **Society & Environment**: Assess societal, legal, and environmental impacts of engineering solutions.
7. **Ethics**: Commit to ethics, human values, diversity, and legal compliance.
8. **Teamwork**: Work effectively as an individual and in multidisciplinary teams.
9. **Communication**: Communicate clearly in reports, presentations, and documentation across diverse groups.
10. **Management & Finance**: Apply management and economic principles in projects and teamwork.

11. **Lifelong Learning**: Engage in continuous learning, adapt to new technologies, and think critically.

Program Specific Outcomes:

PSO1 Design and develop efficient Mechanical systems in real time applications

PSO2 Utilize modern techniques and tools for addressing engineering challenges in the fields of Mechanical Engineering and providing sustainable innovation solutions.

PSO3 Demonstrate research aptitude, work in multidisciplinary teams and implement ethical practices.

Semester–I							
S. No.	**Course Code**	**Course Name**	**Course Type**	**Periods/Week**		**Credits**	**Category**
				L-T-P	**TCP**		
1.	MA25C01	Applied Calculus	T	3-1-0	4	4	BS
2.	ME25C03	Introduction to Mechanical Engineering	T	2-1-0	3	3	ES (PC) – DIC
3.	ME25C01	Engineering Drawing	LIT	2-0-4	6	4	ES (G)
4.	PH25C01	Applied Physics – I	LIT	2-0-2	4	3	BS
5.	CY25C01	Applied Chemistry – I	LIT	2-0-2	4	3	BS
6.	UC25H01	தமிழர்மரபு/Heritage of Tamils	T	1-0-0	1	1	HUM
7.	EN25C01	English Essentials – I	T	2-0-0	2	2	HUM
8.	CS25C02	Computer Programming: Python	LIT	2-0-2	4	3	ES (PC)
9.	ME25C16	Makerspace: Group A – Mechanical Engineering	L	0-0-2	2	1	SD
10.	EE25C11	Makerspace: Group B – EEE & ECE	L	0-0-2	2	1	SD
11.	UC25A01	Life Skills for Engineers – I	---	1-0-2	3	1	HUM
12.	UC25A02	Physical Education – I	---	0-0-4	4	1	HUM
13.		NCC / NSS / NSO	---	---	---	---	---
			Total Credits		**39**	**27**	

Semester – II							
S. No.	**Course Code**	**Course Name**	**Course Type**	**Periods/Week**		**Credits**	**Category**
				L-T-P	**TCP**		
1.	MA25C02	Linear Algebra	T	3-1-0	4	4	BS
2.	ME25C02	Engineering Mechanics	T	3-1-0	4	4	ES (G)
3.	EE25C01	Basic Electrical and Electronics Engineering	T	3-0-0	3	3	ES (G)
4.	PH25C05	Applied Physics (ME) – II	T	2-1-0	3	3	BS
5.	CY25C03	Applied Chemistry (ME) – II	T	2-0-0	2	2	BS
6.	UC25H02	தமிழர்களும் தொழில்நுட்பமும் / Tamils and Technology	T	1-0-0	1	1	HUM
7.	ME25C05	Re-Engineering for Innovation	L	0-0-4	4	2	SD
8.	EN25C02	English Essentials – II	LIT	1-0-2	3	2	HUM
9.	UC25A03	Life Skills for Engineers – II	---	1-0-2	3	1	HUM
10.	UC25A04	Physical Education – II	---	0-0-4	4	1	HUM
11.		Foreign Language^	LIT	1-0-2	3	1	HUM
				Total Credits	**34**	**24**	

^Deutsch / Japanese / Korean

Semester – III							
S. No.	**Course Code**	**Course Name**	**Course Type**	**Periods/Week**		**Credits**	**Category**
				L-T-P	**TCP**		
1.	MA25C09	Computational Differential Equations	T	3-1-0	4	4	BS
2.	ME25C07	Applied Engineering Mechanics	T	3-0-0	3	3	ES (PC)
3.	ME25301	Engineering Thermodynamics	T	4-0-0	4	4	ES (PC)
4.	CE25C11	Strength of Materials	LIT	3-0-2	5	4	ES (PC)
5.	ME25C08	Metallurgy and Materials Science	T	3-0-0	3	3	ES (PC)
6.	EC25C17	Embedded Systems	T	3-0-0	3	3	ES (G)
7.	EN25C03	English Communication Skills Laboratory – I	L	0-0-2	2	1	HUM
8.		Skill Development Course – I	LIT	1-0-2	3	2	SD
				Total Credits	**27**	**24**	

Semester–IV							
S. No.	**Course Code**	**Course Name**	**Course Type**	**Periods/Week**		**Credits**	**Category**
				L-T-P	**TCP**		
1.	CS25C16	Applied Data Science	T	3-0-0	3	3	ES (ET)
2.	ME25C09	Kinematics and Dynamics of Machines	LIT	3-0-2	5	4	ES (PC)
3.	CE25C12	Fluid Mechanics and Machinery	LIT	3-0-2	5	4	ES (PC)
4.	ME25401	Thermal Engineering – I	T	3-0-0	3	3	ES (PC)
5.	ME25402	Manufacturing Processes– I	LIT	3-0-2	5	4	ES (PC)
6.	ME25C10	Standards in Mechanical Engineering	T	1-0-0	1	1	ES (PC)
7.	EN25C04	English Communication Skills Laboratory – II	L	0-0-2	2	1	HUM
8.		Skill Development Course – II	LIT	1-0-2	3	4	SD
Total Credits					**27**	**24**	

Semester–V							
S. No.	**Course Code**	**Course Name**	**Course Type**	**Periods/Week**		**Credits**	**Category**
				L-T- P	**TCP**		
1.		Design of Machine Elements	T	3-0-0	3	3	ES (PC)
2.		Manufacturing Processes – II	LIT	3-0-2	5	4	ES (PC)
3.		Thermal Engineering – II	LIT	3-0-2	5	4	ES (PC)
4.		Measurements and Instrumentation	LIT	2-0-2	4	3	ES (PC)
5.		Programme Elective - I	T	3-0-0	3	3	ES (PE)
6.		Open Elective	T	3-0-0	3	3	---
7.		Skill Development Course – III	LIT	1-0-2	3	2	SD
8.		Industry Oriented Course– I	LIT	1-0-2	3	1	SD
Total Credits					**29**	**23**	
For Honours Degree							
1.		Capstone Design Project-Level I	CDP	0-0-12	12	6	SD
OR							
1.		Honours Elective – I	T	3-0-0	3	3	
2.		Honours Elective – II	T	3-0-0	3	3	
For Minor Degree							
1.		Minor Elective – I	T	3-0-0	3	3	

Semester–V							
S. No.	**Course Code**	**Course Name**	**Course Type**	**Periods/Week**		**Credits**	**Category**
				L-T- P	**TCP**		
2.		Minor Elective – II	T	3-0-0	3	3	

Semester– VI							
S. No.	**Course Code**	**Course Name**	**Course Type**	**Periods/Week**		**Credits**	**Category**
				L-T-P	**TCP**		
1.		Design of Transmission Systems	T	3-1-0	4	4	ES (PC)
2.		Heat and Mass Transfer	LIT	3-0-2	5	4	ES (PC)
3.		FEM applications in Mechanical Engineering	LIT	3-0-2	5	4	ES (PC)
4.		Power Generation Technologies	T	3-0-0	3	3	ES (PC)
5.		Programme Elective - II	T	3-0-0	3	3	ES (PE)
6.		Programme Elective - III	T	3-0-0	3	3	ES (PE)
7.		Computer Aided Modelling Laboratory	L	0-0-4	4	2	ES (PC)
8.		Self-Learning Course	---	---	0	1	---
9.		Industry Oriented Course - II	LIT	1-0-2	3	1	SD
				Total Credits	**30**	**25**	
For Honours Degree							
1.		Capstone Design Project – Level II	CDP	0-0-12	12	6	SD
OR							
1.		Honours Elective – III	T	3-0-0	3	3	
2.		Honours Elective – IV	T	3-0-0	3	3	
For Minor Degree							
1.		Minor Elective – III	T	3-0-0	3	3	
2.		Minor Elective – IV	T	3-0-0	3	3	

Semester– VII							
S. No.	Course Code	Course Name	Course Type	Periods/Week		Credits	Category
				L-T-P	TCP		
1.		Engineering Entrepreneurship Development	LIT	2-0-2	4	3	HUM
2.		Climate Change and Sustainability	T	2-0-0	2	2	HUM
3.		Mechatronics and Automation	LIT	3-0-2	5	4	ES (PC)
4.		Programme Elective - IV	T	3-0-0	3	3	ES (PE)
5.		Programme Elective - V	T	3-0-0	3	3	ES (PE)
6.		Project Management	T	2-0-0	2	2	HUM
7.		Industrial Training	---	---	---	1	SD
Total Credits					**19**	**18**	
For Honours Degree							
1.		Capstone Design Project – Leve	CDP	0-0-12	12	6	SD
OR							
1.		Honours Elective – V	T	3-0-0	3	3	
2.		Honours Elective – VI	T	3-0-0	3	3	
For Minor Degree							
1.		Minor Elective – V	T	3-0-0	3	3	
2.		Minor Elective – VI	T	3-0-0	3	3	

Semester– VIII							
S. No.	Course Code	Course Name	Course Type	Periods/Week		Credits	Category
				L-T-P	TCP		
1.		Project Work / Internship cum Project Work	PW / IPW	0-0-16	16	8	SD
Total Credits					**16**	**8**	

Total Number of Credits: 173

PROGRAMME ELECTIVE COURSES – STREAMS

Thermal	Manufacturing	Design	Materials	Industrial & Systems Engineering
Refrigeration &HVAC Systems	Digital Manufacturing	Computer-Aided Design and Prototyping	Composite Materials	Optimisation Techniques
Battery Technology	Robotic Systems	Tool Design	Fundamentals of Nanomaterials	Lean Manufacturing
Gas Dynamics and Jet Propulsion	Industrial IoT	Tribological Design	Nano Technology and Surface Engineering	Supply Chain Analytics
Vehicle Technology	Welding Engineering	Mechanical Vibrations	Materials for Electronics	Quality Engineering
Computational Fluid Dynamics	Design for Manufacturing and Assembly	Digital Twin	Materials Characterization	Life Cycle Assessment
Renewable Energy Systems	Additive Manufacturing	Electric Vehicle Design	Mechanical Behaviour of Materials	---
Energy Conservation and Management	-	Integrated Product Design and Development	Smart Materials	-

Semester I

MA25C01	Applied Calculus	L	T	P	C
		3	1	0	4

Course Objectives:

- To provide technical competence of modelling engineering problems using calculus.
- To apply the calculus concepts in solving engineering problems using analytical methods and computational tools.

Differential Calculus: Functions, graph of functions, New functions from old functions, Limit of a function, Continuity, Limits at infinity, Derivative as a function, Maxima and Minima of functions of single variable, Mean value theorem, Effect of derivatives on the shape of a graph.

Activities: Visualization of the functions, Maxima and Minima of a function using open-source software, Solving of Competitive Examination questions (Ex. GATE).

Functions of Several Variables: Partial derivatives, Chain rule, Total derivative, Maxima and minima of functions of two variables, Method of Lagrange's Multipliers, Application problems in engineering.

Activities: Partial Derivatives with two or three variables, Maxima and Minima of a function using open-source software, Solving of Competitive Examination questions (Ex. GATE).

Integral Calculus: Fundamental theorem of Calculus, Indefinite integrals and the Net Change Theorem, Improper integrals, Arc Length, Area of Region, Area of surface of revolution.

Activities: Definite and Indefinite Integrals, Determination of Area, Solving of Competitive Examination questions (Ex. GATE).

Multiple Integrals: Iterated integrals and Fubini's theorem, Evaluation of double integrals, change of order of integration, change of variables between Cartesian and polar co-ordinates, evaluation of triple integrals-change of variables between Cartesian and cylindrical and spherical co-ordinates.

Activities: Double integrals and triple integrals using open-source software, Solving of Competitive Examination questions (Ex. GATE).

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%.

Assessment Methodology: Assignments (20%), Solution to application-oriented problems using software (20%), Solving of GATE questions (20%), Internal Examinations (40%).

References:

1. Anton, H., Bivens, I. C., & Davis, S. (2021). Calculus: Early transcendentals. John Wiley & Sons.
2. Ron Larson and David C. Falvo,(2013), Calculus: an Applied Approach. Cengage

Learning.
3. Stewart, J., Clegg, D., & Watson, S. (2019). *Calculus: Early transcendentals*.
4. Thomas, G. B., Jr., Weir, M. D., Hass, J., & Heil, C. (2018). *Thomas' calculus: Early transcendentals*. Pearson.
5. Singh, K. (2019). *Engineering mathematics through applications*. Bloomsbury Publishing.
6. Grewal, B. S. (2012). *Higher engineering mathematics.* Khanna Publishers.

E-resources:

1. https://math.libretexts.org/Bookshelves/Calculus/Map%3A_Calculus__Early_Transcendentals_(Stewart)/
2. https://openstax.org/books/calculus-volume-1/
3. https://tutorial.math.lamar.edu/Classes/CalcII/CalcII.aspx
4. SCILAB, https://www.scilab.org/

	Description of CO	**PO**	**PSO**
CO1	Explain the meaning of derivative, integral, and their geometric and physical interpretations.	---	---
CO2	Apply differentiation and integration techniques to compute maxima, minima, and area.	PO1(3)	PSO1(2) PSO2(2)
CO3	Analyze the behavior of single and multivariable functions using derivatives and partial derivatives.	PO2(3)	PSO1(2) PSO3(1)
CO4	Utilize modern computational software and online platforms to deepen understanding, perform complex calculations, and visualize mathematical concepts.	PO5(2) PO11(1)	PSO2(3) PSO3(1)

ME25C03	Introduction to Mechanical Engineering	L	T	P	C
		2	1	0	3

Course Objectives:

- To impart the fundamental concepts and principles of various fields such as Manufacturing, Materials, Mechanics, thermal engineering in Mechanical Engineering.

Engineering: History and evolution of mechanical engineering, Basic mechanical engineering principles (force, motion, energy, work, power), Units and dimensions, SI system, Ethics and professionalism in engineering.

Activities: Interactive quiz, Conversion between SI and other unit systems.

Mechanics of Materials and Structures: Stress and strain, types of stresses (tensile, compressive, shear), Elasticity and plasticity, Mechanical properties of materials (strength, toughness, hardness), Introduction to bending, torsion, and axial loading, Simple structural analysis and design concepts.

Activities: Demonstration of Simple truss or beam problems solved using software.

Energy Interactions: System, Energy Transfer, Conduction, convection, and radiation, Working principle of Heat Engines, Refrigeration and HVAC systems.

Activities: Demonstration of working model of internal combustion engine & refrigerator, Virtual demonstration of Thermodynamic cycles.

Machine Elements: Gears, bearings, shafts, fasteners, couplings, Selection of machine components, Quality control and safety in mechanical engineering.

Activities: Demonstration of working of Gears, bearings, etc. in a mechanical system.

Manufacturing Processes: Casting, forming, machining & joining processes, CNC and additive manufacturing, overview of smart manufacturing.

Activities: Demonstration of various machining processes, 3D printing of simple parts.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (10%), Assignments (40%) and Internal Examinations (50%)

References:

1. Wickert, J., & Lewis, K. (2016). An Introduction to Mechanical Engineering. Cengage Learning.
2. Rajput, R. K., (2017). Fundamentals of Mechanical Engineering, Laxmi Publications.

E-resources:

1. MIT OpenCourseWare – Mechanical Engineering https://ocw.mit.edu
2. PhET Simulations – University of Colorado Boulder https://phet.colorado.edu
3. LibreTexts Engineering https://eng.libretexts.org

	CO Description	PO	PSO
CO1	Explain core mechanical engineering concepts.	---	---
CO2	Apply basic engineering calculations in mechanical systems.	PO1(3)	PSO1(3) PSO2(2)
CO3	Identify common manufacturing processes for engineering applications.	PO2(2)	PSO3(2)

ME25C01	Engineering Drawing	L	T	P	C
		2	0	4	4

Course Objectives:

- To impart knowledge on dimensions and drawing standards.
- To explore the orthographic projection of lines and solids.
- To provide the understanding of orthographic, isometric and perspective views.

Fundamentals: Drawing instruments, Drawing standards (BIS), Lettering in engineering, Sheet layout, elements of dimensioning, Systems of dimensioning. Free hand sketching of 2D & 3D objects, Conics – Ellipse, Parabola and Hyperbola.

Activities: Virtual Demonstration of Conics and Cycloids.

Orthographic Projection: First angle projection, Projection of points, straight lines and planes.

Projection of Solids: Simple Solids, Section of Solids, Development of Surfaces

Activities: Development of models of various solids and virtual demonstration of sectioning, CAD modelling of 2D objects.

Isometric Projection: Isometric Scale, Projection of Simple solids.

Activities: Conversion of 3D into 2D orthographic views, CAD modelling of 3D objects.

Perspective Projection: Simple solids projection

Activities: Virtual demonstration of perspective views.

Project: Development of 2D objects and 3D objects using CAD tools.

Weightage:
Continuous Assessment: 50%
End Semester Examinations: 50%

Assessment Methodology:
Project – 10%, Models - 5%, Assignments - 35% and Internal Examinations - 50%

References:

1. Natarajan, K. V. (2025). A Text Book of Engineering Graphics. Dhanalakshmi Publisher.
2. Venugopal, K., & Prabhu Raja, V. (2022). Engineering Drawing + AutoCAD. New Age International Publishers.

E-Resources:

1. CAD Software – https://www.freecadweb.org/
2. Engineering Drawing and Computer Graphics, Prof. Rajaram Lakkaraju (IIT Kharagpur) – https://onlinecourses.nptel.ac.in/noc22_me105/preview
3. MIT Design Handbook: Engineering Drawing and Sketching – https://ocw.mit.edu/courses/2-007-design-and-manufacturing-i-spring-2009/pages/related-resources/drawing_and_sketching/

	CO Description	PO	PSO1
CO1	Explain the advantages of engineering drawing in engineering applications	---	
CO2	Apply the concepts of projections in formulating various solid parts in engineering systems.	PO1(3)	PSO1(2)
CO3	Analyse the various view and interpret the engineering drawings.	PO2(3)	PSO1(2)
CO4	Use CAD tools for creation of various models.	PO3(1)	PSO2(2)
CO5	Critically think and develop innovative models.	PO11(1)	PSO3(1)

PH25C01	Applied Physics – I	L	T	P	C
		2	0	2	3

Course Objective(s):

- To impart knowledge and expose the essentials of physics in various engineering applications.

Properties of Matter: Elasticity, Cantilever, Young's modulus (non-uniform bending), Girders: Bridges and buildings, Viscosity: Stokes method, Surface tension: drop weight method, Thermal expansion, Thermal stress, Bimetallic strips, Expansion joints

Practical: Non-Uniform bending, Young's modulus of the material, Torsional pendulum, Rigidity modulus of the wire and moment of inertia of the disc.

Activities: Virtual demonstration of thermal stress.

Oscillations: Simple Harmonic motion, Torsional pendulum, Couple per unit twist, Damped and Forced Oscillation

Waves: Waves on a stretched string, Energy and Power, standing waves, Ultrasonics, piezo, electric method, Acoustic grating, Electromagnetic waves: Maxwell equation, Production of EM waves by dipole antenna, Propagation of EM waves in free space , wave equation, Cell phone reception

Practical: Melde's string experiment – Frequency of an electrically vibrating metal tip.

Activities: Virtual demonstration of propagation of EM waves

Quantum Mechanics: Black body radiation, Photoelectric effect, de Broglie hypothesis, Schrodinger Wave equation, Particle in a box (infinite potential well - three-dimensional box), Barrier penetration and quantum tunnelling.

Practical: Photo-electric effect, Determination of Planck's constant.

Activities: Virtual demonstration of Scanning Transmission Electron Microscope

Applied Optics: Interference: Air wedge, Michelson's Interferometer, Fiber optics: Structure of a fiber, Fiber Optic Communication System, Fiber Sensors (Virtual demo), Displacement, pressure sensor and Temperature sensor, Einstein Co-efficient, Nd:YAG laser, CO_2 laser (construction, functioning and applications), dye laser

Practical: Ruling width of Compact disc using Laser, Thickness of a thin sheet/wire using Air wedge Method.

Activities: Demonstration of sensors and applications of Lasers

Weightage: Continuous Assessment: 50%, End Semester Examinations: 50%

Assessment Methodology: Quiz (5%), Assignments (20%), Flipped Class (5%), Practical (30%), Internal Examinations (40%)

References:

1. Young, H. D., & Freedman, R. A. (2020). University physics with modern physics. Pearson.
2. Gaur, R. K., & Gupta, S. L. (2022). Engineering physics. Dhanpat Rai Publications.
3. Mathur, D. S. (2010). Elements of properties of matter. S. Chand Publishing.
4. Griffiths, D. J. (2018). Introduction to quantum mechanics. Cambridge University Press.
5. Silfvast, W. T. (2008). Laser fundamentals (2nd ed.). Cambridge University Press.

E-Resources:

1. Barrier penetration problem and Quantum tunnelling: https://archive.nptel.ac.in/courses/115/104/115104096/
2. EM waves and wireless channelling: https://onlinecourses.nptel.ac.in/noc24_ee31/preview
3. CO2 Laser: https://onlinecourses.nptel.ac.in/noc25_ph03/preview
4. Bimetallic Strips _ https://www.youtube.com/watch?v=WZQ8lvxdzDk
5. Cell phone Reception_ https://www.youtube.com/watch?v=1JZG9x_VOwA
6. Dipole Antenna_ https://www.youtube.com/watch?v=4xF1Fq2wB1I
7. Optical Sensors_ https://auece.digimat.in/nptel/courses/video/108106173/L02.html
8. Scanning Tunnelling Electron Microscope_ https://www.youtube.com/watch?v=XNYZYbXNWQA

	Description of CO	**PO**	**PSO**
CO1	Explain the physics concepts in various applications.	---	---
CO2	Apply the principles of wave optics and laser physics in practical systems.	PO1(3)	PSO1(2) PSO2(2)
CO3	Analyse the behaviour of materials under different conditions.	PO2(2)	PSO1(2) PSO3(1)
CO4	Conduct experiments in groups and interpret the data.	PO4(3) PO8(1)	PSO1(2) PSO2(2)

CY25C01	Applied Chemistry – I	L	T	P	C
		2	0	2	3

Course Objectives:

- To provide students with a solid understanding of the chemical principles for engineering applications.
- To introduce the chemical properties of materials and how these properties influence the selection and use of materials in engineering systems.
- To impart practical applications of chemistry in commonly used engineering devices

Water Technology: Water quality parameters and standards. Industrial feed water, Remediation. Municipal water treatment. Desalination.

Practical: Analysis of alkalinity, hardness and dissolved oxygen.

Activity: Coagulation of water sample using Alum

Nano-chemistry: Classification, Size, dependent properties. Preparation of nanomaterials, Top-down and Bottom-Up approaches, Applications (Flipped classroom).

Practical: Preparation of nanoparticles by Sol-Gel method.

Electrochemistry: Electrochemical cell, Electrode potential., Redox reaction. Conductivity of electrolytes, Factors.

Practical: Conductometric titrations

Activity: Electrochemical cell demonstration

Corrosion & Control: Chemical and electrochemical corrosions, galvanic series, factors influencing corrosion, Electrochemical protection. Organic and Inorganic coating.

Practical:

- Corrosion study by weight loss and salt spray method.
- Potentiometry/UV-visible spectrophotometer.

Activities: Case Study on Corrosion in Pipelines and Electronics, Control measures for a corroded metal

Batteries: Conventional, Contemporary and Emerging battery storage technologies, Primary & Secondary Batteries, Battery Pack, Battery Materials, Performance Parameters, Testing, Safety aspects.

Practical: Measurement of EMF, Internal Resistance, Charge and Discharge Characteristics.

Activities: Demonstration of battery pack in e-vehicles.

Weightage: Continuous Assessment: 50%, End Semester Examinations: 50%

Assessment Methodology: Quiz (5%), Assignments (20%), Flipped Class (5%), Practical (30%), Internal Examinations (40%)

References:

1. Jain, P. C., & Jain, M. (2015). Engineering Chemistry (17th ed.). Dhanpat Rai Publishing Company (P) Ltd.
2. Dara, S. S. (2004). A Textbook of Engineering Chemistry. Chand Publications.
3. Sachdeva, M. V. (2011). Basics of Nano Chemistry. Anmol Publications Pvt Ltd.
4. Friedrich, E. (2014). Engineering Chemistry. Medtech.

E-Resources:

1. Water and Wastewater Engineering (Prof. Ligy Philip, IIT Madras) – https://nptel.ac.in/courses/105106202.
2. Electrochemical Energy Systems (Prof. S. Mitra, IIT Madras) – https://nptel.ac.in/courses/113106028.
3. Corrosion (Prof. Kallol Mondal, IIT Kanpur) – https://nptel.ac.in/courses/112104088
4. Chemistry of Battery Systems (Prof. V. R. Marathe, IIT Madras) – https://nptel.ac.in/courses/115106130
5. Resource on all battery types, testing, and safety – https://batteryuniversity.com/articles

	Description of CO	**PO**	**PSO**
CO1	Understand the importance of chemistry applications with underlying mechanisms.	---	
CO2	Apply the chemistry concepts in widely used devices.	PO1(3)	PSO1(2) PSO2(2)
CO3	Analyse the effect of various chemical parameters on performance of engineering systems.	PO2(2)	PSO1(2) PSO2(1)
CO4	Perform experimentations as a group and interpret the results.	PO4(3) PO8(1)	PSO2(2) PSO3(2)
CO5	Communicate findings through case studies and reports	PO9(1)	PSO2(2) PSO3(3)

UC25H01	தமிழர் மரபு	L	T	P	C
		1	0	0	1

மொழி மற்றும் இலக்கியம்: இந்திய மொழிக் குடும்பங்கள், திராவிட மொழிகள், தமிழ் ஒரு செம்மொழி, தமிழ் செவ்விலக்கியங்கள், சங்க இலக்கியத்தின் சமயச் சார்பற்ற தன்மை, சங்க இலக்கியத்தில் பகிர்தல் அறம், திருக்குறளில் மேலாண்மைக் கருத்துக்கள், தமிழ்க் காப்பியங்கள், தமிழகத்தில் சமண பௌத்த சமயங்களின் தாக்கம், பக்தி இலக்கியம், ஆழ்வார்கள் மற்றும் நாயன்மார்கள், சிற்றிலக்கியங்கள், தமிழில் நவீன இலக்கியத்தின் வளர்ச்சி, தமிழ் இலக்கிய வளர்ச்சியில் பாரதியார் மற்றும் பாரதிதாசன் ஆகியோரின் பங்களிப்பு.

மரபு – பாறை ஓவியங்கள் முதல் நவீன ஓவியங்கள் வரை – சிற்பக் கலை:நடுகல் முதல் நவீன சிற்பங்கள் வரை, ஐம்பொன் சிலைகள், பழங்குடியினர் மற்றும் அவர்கள் தயாரிக்கும் கைவினைப் பொருட்கள், பொம்மைகள் , தேர் செய்யும் கலை, சுடுமண் சிற்பங்கள், நாட்டுப்புறத் தெய்வங்கள், குமரிமுனையில் திருவள்ளுவர் சிலை, இசைக் கருவிகள், மிருதங்கம், பறை, வீணை, யாழ், நாதஸ்வரம், தமிழர்களின் சமூக பொருளாதார வாழ்வில் கோவில்களின் பங்கு.

நாட்டுப்புறக் கலைகள் மற்றும் வீர விளையாட்டுகள்: தெருக்கூத்து, கரகாட்டம், வில்லுப்பாட்டு, கணியான் கூத்து, ஒயிலாட்டம், தோல்பாவைக் கூத்து, சிலம்பாட்டம், வளரி, புலியாட்டம், தமிழர்களின் விளையாட்டுகள்.

தமிழர்களின் திணைக் கோட்பாடுகள்: தமிழகத்தின் தாவரங்களும், விலங்குகளும், தொல்காப்பியம் மற்றும் சங்க இலக்கியத்தில் அகம் மற்றும் புறக் கோட்பாடுகள், தமிழர்கள் போற்றிய அறக்கோட்பாடு, சங்ககாலத்தில் தமிழகத்தில் எழுத்தறிவும், கல்வியும், சங்ககால நகரங்களும் துறை முகங்களும், சங்ககாலத்தில் ஏற்றுமதி மற்றும் இறக்குமதி, கடல்கடந்த நாடுகளில் சோழர்களின் வெற்றி.

இந்திய தேசிய இயக்கம் மற்றும் இந்திய பண்பாட்டிற்குத் தமிழர்களின் பங்களிப்பு: இந்திய விடுதலைப்போரில் தமிழர்களின் பங்கு, இந்தியாவின் பிறப்பகுதிகளில் தமிழ்ப் பண்பாட்டின் தாக்கம் , சுயமரியாதை இயக்கம் இந்திய மருத்துவத்தில், சித்த மருத்துவத்தின் பங்கு, கல்வெட்டுகள், கையெழுத்துப்படிகள், தமிழ்ப் புத்தகங்களின் அச்சு வரலாறு.

References:

1. தமிழக வரலாறு, மக்களும் பண்பாடும், கே.கே. பிள்ளை (வெளியீடு: தமிழ்நாடு பாடநூல் மற்றும் கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ், முனைவர் இல. சுந்தரம். (விகடன் பிரசுரம்).
3. கீழடி, வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. **பொருநை, ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)**
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils - The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.)
9. Keeladi - 'Sangam City C ivilization on the banks of river Vaigai' (Jointly Published by: Department ofArchaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Publishedby: The Author)
11. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.

UC25H01	Heritage of Tamils	L	T	P	C
		1	0	0	1

Language and Literature: Language Families in India, Dravidian Languages, Tamil as a Classical Language, Classical Literature in Tamil, Secular Nature of Sangam Literature, Distributive Justice in Sangam Literature, Management Principles in Thirukural, Tamil Epics and Impact of Buddhism & Jainism in Tamil Land, Bakthi Literature Azhwars and Nayanmars, Forms of minor Poetry - Development of Modern literature in Tamil - Contribution of Bharathiyar and Bharathidhasan.

Heritage - Rock Art Paintings to Modern Art – Sculpture: Hero stone to modern sculpture, Bronze icons, Tribes and their handicrafts, Art of temple car making, Massive Terracotta sculptures, Village deities, Thiruvalluvar Statue at Kanyakumari, Making of musical instruments, Mridhangam, Parai, Veenai, Yazh and Nadhaswaram, Role of Temples in Social and Economic Life of Tamils.

Folk and Martial Arts: Therukoothu, Karagattam, Villu Pattu, Kaniyan Koothu, Oyillattam, Leather puppetry, Silambattam, Valari, Tiger dance, Sports and Games of Tamils.

Thinai Concept of Tamils: Flora and Fauna of Tamils & Aham and Puram Concept from Tholkappiyam and Sangam Literature, Aram Concept of Tamils, Education and Literacy during Sangam Age, Ancient Cities and Ports of Sangam Age, Export and Import during Sangam Age, Overseas Conquest of Cholas.

Contribution of Tamils to Indian National Movement and Indian Culture:
Contribution of Tamils to Indian Freedom Struggle, The Cultural Influence of Tamils over the other parts of India, Self-Respect Movement, Role of Siddha Medicine in Indigenous Systems of Medicine, Inscriptions & Manuscripts, Print History of Tamil Books

References:

1. தமிழக வரலாறு, மக்களும் பண்பாடும், கே.கே. பிள்ளை (வெளியீடு: தமிழ்நாடு பாடநூல் மற்றும், கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ், முனைவர் இல. சுந்தரம். (விகடன் பிரசுரம்).
3. கீழடி, வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. பொருநை, ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils, The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
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8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.)
9. Keeladi, 'Sangam City C ivilization on the banks of river Vaigai' (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Publishedby: The Author)
11. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Bookand Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL), Reference Book.

EN25C01	English Essentials – I	L	T	P	C
		2	0	0	2

Course Objectives:

- Enhance learners' listening and speaking skills to understand and deliver speeches effectively
- Equip students with the skills to write clear, coherent, and grammatically correct texts for various purposes.
- Strengthen the ability to comprehend, interpret, and analyse written English across diverse contexts.

Speaking Skills: Self-Introduction (Tenses, Adjectives) Expressing opinions (Subject-Verb Agreement), Participating in Conversations (Speech Acts - agreeing & disagreeing – synonyms and antonyms**)**

Suggested Activities: Self-Introduction, Just a Minute (JAM) Video recording, Situational role plays, Spell Bee, Word Substitution, Usage of Apps.

Listening Skills: Listening to Simple Conversations (Understanding tone and intent), Short Speeches / Stories, Extracting information, Pronunciation, Listening to Various Accents.

Suggested Activities: Listening and Repeating, Gap fill exercises, Note-taking

Reading Skills: Reading Strategies – (Skimming, scanning, predicting) intensive reading - short passages and long passages on suggested themes (Sentence Patterns, Prefixes and suffixes, idioms and phrases).

Activities: Reading - newspaper and digital articles, Cloze, Reading comprehension, note making and summarising,

Writing Skills: Word Substitution, Sentence Formation, Hints Development (Guided Writing), Writing Different Types of Paragraphs - (Sentence Structure) – Letter Writing / Emails (Informal)

Activities: Error Detection, Picture and poster description, Descriptive, Narrative and Comparative paragraphs, Brainstorming and Mind Mapping - Informal letters/ Emails

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (10%), Assignments (20%), Speaking Task (10%), Reading Task (10%), Writing Task (10%), Internal Examinations (40%).

References:

1. Miller, K. Q., & Wahl, S. T. (2023). *Business and Professional Communication: KEYS for Workplace Excellence* (5th ed.). SAGE Publications.
2. Kumar, Sanjay & Pushpalatha. (2018). *English Language and Communication Skills for Engineers*. India: Oxford University Press.
3. Sharma, S., & Mishra, B. (2024). *Communication Skills for Engineers and Scientists* (2nd ed.). PHI Learning.

E-resources:

1. Cambridge English – https://www.cambridgeenglish.org/learning-english/grammar-and-vocabulary/
2. Perfect English Grammar – https://www.perfect-english-grammar.com/
3. British Council – Learn English - https://learnenglish.britishcouncil.org/grammar
4. Speechling – https://speechling.com/

5. mePro by Pearson – https://mepro.pearson.com/
6. TED Talks – https://www.ted.com/

	Description of CO	**PO**	**PSO**
CO1	Listen and comprehend spoken English, take and draft notes.	---	---
CO2	Apply vocabulary and grammar appropriately to communicate in written and spoken forms.	PO1(3)	PSO1(2) PSO3(3)
CO3	Analyze texts in different contexts using appropriate reading strategies.	PO2(2)	PSO2(1)
CO4	Communicate thoughts and ideas in real life situations.	PO9(2)	PSO3(2)
CO5	Develop communication skills relevant to engineering and technology.	PO11(1)	PSO3(3)

CS25C02	Computer Programming: Python	L	T	P	C
		2	0	2	3

Course Objectives:

- To equip engineering students with the foundational knowledge and practical skills in Python programming to analyse and solve computational problems effectively.
- To foster problem-solving, critical thinking, and modular programming skills essential for engineering domains.

Introduction to Python: Problem Solving, Problem Analysis Chart, Developing an Algorithm, Flowchart and Pseudocode, Interactive and Script Mode, Indentation, Comments, Error messages, Variables, Reserved Words, Data Types, Arithmetic operators and expressions, Built-in Functions, Importing from Packages.

Practical: Problem Analysis Chart, Flowchart and Pseudocode Practices. (Minimum three)

Control Structures: if, if-else, nested if, multi-way if-elif statements, while loop, for loop, nested loops, pass statements.

Practical: Usage of conditional logics in programs. (Minimum three)

Functions: Hiding redundancy, complexity; Parameters, arguments and return values; formal vs actual arguments, named arguments, Recursive & Lambda Functions.

Practical: Usage of functions in programs. (Minimum three)

Strings & Collections: String Comparison, Formatting, Slicing, Splitting, Stripping, Lists, tuples, and dictionaries, basic list operators, searching and sorting lists; dictionary literals, adding and removing keys, accessing and replacing values.

Practical: String manipulations and operations on lists, tuples, sets, and dictionaries. (Minimum three)

File Operations: Create, Open, Read, Write, Append and Close files. Manipulating directories, OS and Sys modules, reading/writing text and numbers, from/to a file; creating and reading a formatted file (csv, tab-separated, etc.).

Practical: Opening, closing, reading and writing in formatted file format and sort data. (Minimum three)

Packages: Built-in modules, User-Defined modules, Numpy, SciPy, Pandas, Scikit-learn**.**

Practical: Usage of modules and packages to solve problems. (Minimum three), Project (Minimum Two)

Weightage: Continuous Assessment: 50%, End Semester Examinations: 50%

Assessment Methodology: Quiz (5%), Project (15%), Assignment Programs (25%), Practical (25%), Internal Examinations (30%)

References:

1. Matthes, E. (2019). *Python crash course: A hands-on, project-based introduction to programming* (2nd ed.). No Starch Press.
2. Brown, M. C. (2018). *Python: The complete reference* (4th ed.). McGraw Hill Publishers.
3. Guttag, J. V. (2016). *Introduction to computation and programming using Python: With applications to understanding data* (2nd ed.). MIT Press.
4. McKinney, W. (2017). *Python for data analysis: Data wrangling with pandas, NumPy, and IPython*. Shroff/O'Reilly.

E-Resources:

1. Official Python Documentation – https://docs.python.org/3/
2. Python Tutorials – https://www.w3schools.com/python/
3. NumPy – https://numpy.org/doc/
4. SciPy – https://scipy.org/
5. Google's Python class – https://developers.google.com/edu/python/

	Description of CO	**PO**	**PSO**
CO1	Explain the potential usage of Python in engineering applications	---	---
CO2	To apply the concepts of Python in solving engineering problems and formulate new projects.	PO1 (2) PO5 (2)	PSO2(2) PSO3(1)
CO3	To interpret the data and effectively communicate in groups.	PO2 (3) PO8 (1) PO9 (1)	PSO3(1) PSO3(2)
CO4	Adapt new programming concepts and technologies in the profession.	PO11 (1)	PSO2(2)

ME25C16	Makerspace : Group A– Mechanical Engineering	L	T	P	C
		0	0	2	1

Course Objectives:

1. To develop practical skills in the **selection, handling and safe use of hand tools and power tools** for assembly, disassembly and basic fabrication activities.
2. To provide hands-on experience in **welding processes, joint preparation and fabrication of simple components** using appropriate materials and welding techniques.
3. To introduce students to **contemporary digital manufacturing practices**, including solid modelling, assembly modelling, 3D printing and laser cutting.

List of Activities:

(A). Dis-assembly & Assembly Practices

i. Tools and its handling techniques.
ii. Dis-assembly and assembly of home appliances – Grinder Mixer Grinder, Ceiling Fan, Table Fan & Washing Machine.
iii. Dis-assembly and assembly of Air-Conditioners & Refrigerators.
iv. Dis-assembly and assembly of a Bicycle.

(B). Welding Practices

i. Welding Procedure, Selection & Safety Measures.
ii. Power source of Arc Welding – Gas Metal Arc Welding & Gas Tungsten Arc Welding processes.
iii. Hands-on session of preparing base material & Joint groove for welding.
iv. Hands-on session of MAW, GMAW, GTAW, on Carbon Steel & Stainless Stell plates / pipes, for fabrication of a simple part.

(C). Contemporary Digital Manufacturing Practices

i. Demonstration of Solid Modelling of components.
ii. Demonstration of Assembly Modelling of components.
iii. Fabrication of simple components / parts using 3D Printers.
iv. Demonstration of cutting of wood / metal in different complex shapes using Laser Cutting Machine.

References:

1. Stephen Christena, Learn to Weld: Beginning MIG Welding and Metal Fabrication Basics, Crestline Books, 2014.
2. H. Lipson, Fabricated - The New World of 3D Printing, Wiley, 1st edition, 2013.
3. Refrigeration and Air Conditioning C. P. Arora 3rd McGraw Hill Education 2014
4. Welding: Principles and Applications Larry Jeffus 9th Cengage Learning 2020/2021.
5. Computer Aided Design and Manufacturing M. M. M. Sarcar, K. Mallikarjuna Rao & K. Lalit Narayan 2nd PHI Learning 2012
6. Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing Ian Gibson, David Rosen, Brent Stucker & Mahyar Khorasani 3rd Springer 2021.

	Description of CO	PO	PSO
CO1	Demonstrate the proper selection, use and handling of basic hand tools and power tools for assembly and disassembly activities.	PO (3)	---
CO2	Apply appropriate welding processes and safety practices to prepare joints and fabricate simple mechanical components.	PO1(3)	PSO2(1)
CO3	Develop three-dimensional solid and assembly models of simple mechanical components using appropriate CAD software.	PO5(3)	PSO2(2)
CO4	Fabricate simple components using contemporary manufacturing processes such as 3D printing and laser cutting while adhering to applicable safety practices.	PO8(2)	PSO3(2)

EE25C11	Makerspace : Group B – EEE & ECE	L	T	P	C
		0	0	2	1

Course Objectives:

1. To develop practical skills in the selection, handling, assembly and disassembly of electrical and electronic components using appropriate tools and techniques.
2. To provide hands-on experience in electrical wiring, soldering, electronic circuits, and the integration of sensors, actuators and microcontrollers for basic applications.

List of Activities:

(A) Electrical Wiring Practices

i. Electrical Installation tools, equipment & safety measures.
ii. Hands-on session of basic electrical connections for Fuses, Miniature Circuit Breakers and Distribution Box.
iii. Hands-on session of electrical connections for Lightings, Fans, Calling Bells.
iv. Hands-on session of electrical connections for Motors & Uninterruptible Power Supply.

(B) Electronics Components / Equipment Practices

i. Electronic components, equipment & safety measures.
ii. Dis-assembly and assembly of Computers.
iii. Hands-on session of Soldering Practices in a Printed Circuit Board.
iv. Hands-on session of Bridge Rectifier, Op-Amp and Transimpedance amplifier.
v. Hands-on session of integration of sensors and actuators with a Microcontroller.
vi. Demonstration of Programmable Logic Control Circuit.

References:

1. Code of Practice for Electrical Wiring Installations (IS 732:2019)
2. Electrical Wiring, Estimating & Costing — S. L. Uppal, Sixth Edition, Khanna Publishers,1987.
3. Electronic Devices and Circuit Theory — Boylestad & Nashelsky,11th Edition, Pearson Education India, 2015.
4. Practical Electronics for Inventors — Scherz & Monk, 4th Edition, MGH, 2016.
5. Programmable Logic Controllers — Frank D. Petruzella,6th Edition, MGH, 2023.

	Description of CO	PO	PSO
CO1	Demonstrate the proper selection, use and handling of electrical installation tools and equipment for basic electrical wiring practices.	---	---
CO2	Carry out basic electrical wiring connections for domestic and low-power applications, applying appropriate safety and protection measures.	PO1(3)	PSO2(1)
CO3	Assemble, disassemble and test basic electronic components and circuits using appropriate soldering and testing techniques.	PO5(2)	PSO2(2)
CO3	Integrate sensors and actuators with a microcontroller and demonstrate basic programmable control applications.	PO11(2)	PSO3(2)

UC25A01	Life Skills for Engineers – I	L	T	P	C
		1	0	2	1

Course Objectives

- To equip engineering students with essential life skills encompassing personal and emotional development, effective management of time and stress, financial literacy, digital safety, and civic responsibility.
- To enhance self-awareness, interpersonal skills, and resilience to prepare students for the professional and personal challenges of engineering careers and life beyond academics**.**

Personal and Emotional Development: Self-Awareness & Personality, Emotional Intelligence & Empathy, Positive thinking, Right attitude, Stress & Anger Management, Goal-Setting & Time Management, Growth Mindset & Resilience.

Activities: Personality tests (MBTI, DISC), reflection journals, Empathy circle, role-playing difficult conversations, Guided mindfulness sessions, stress relief toolkit creation, Vision board creation, weekly time audit and planner, Group challenge scenarios, resilience journal.

Management Skills: Financial Literacy: Budgeting & Saving, Nutrition, Health, and Hygiene, Digital Literacy & Online Safety, Civic Responsibility & Ethics

Activities: Create a monthly budget, financial simulation game, Meal planning workshop, physical wellness challenge, Social media audit, privacy and safety scenarios, Community service, values debate.

Weightage: Continuous Assessment: 100%

Assessment Methodology: Assignments (20%), Flipped Class & Worksheets (10%), Practical (30%), Internal Examinations (40%)

References:

1. Khera, S. (2003). *You can win*. Macmillan.
2. Levesque, H. (n.d.). *Life skills 101: A practical guide to leaving home and living on your own*. (Publication year not specified)
3. Mitra, B. K. (2017). *Personality development & soft skills* (3rd impression). Oxford University Press.
4. ICT Academy of Kerala. (2016). *Life skills for engineers*. McGraw Hill Education (India) Private Ltd.

	Description of CO	PO	PSO
CO1	Understand personality traits and emotional intelligence, in interpersonal interactions.	---	---
CO2	To work and execute as a team through successful implementation of set goals.	PO7 (1) PO8 (2) PO9 (2)	PSO3(2)
CO3	Develop and implement best practices in day-to-day life, in terms of planning and execution.	PO11 (3)	PSO3(2)

UC25A02	Physical Education - I	L	T	P	C
		0	0	4	1

Course Objectives:

- To impart the fundamentals of physical education for development of students' physical, mental, and social well-being.
- To instill a lifelong appreciation for physical activity towards the development of positive attitude and fostering values of team work and sportsmanship.

Introduction to physical education: Exercise for Good Posture – Conditioning and Calisthenics for Before start, Jogging, Bending, Twisting, Standing, Sitting and Relaxation, Training on First Aid practices.

Participation of athletic events: Rules and regulations of important athletic events, Sprint, Jumps, Throws and Hurdles.

Skill development in any one of the following outdoor games: Basket Ball, Volley Ball, Ball Badminton, Football, Hockey, Kho-Kho, Kabaddi, Cricket, Hand ball and Tennis.

Skill development in any one of the following indoor games: Shuttle Badminton, Chess and Table Tennis.

Weightage: Continuous Assessment: 100%

Assessment Methodology: Attendance (60%), Quiz (10%), Participation in Sports and Games (20%) and Viva Voce (10%)

References:

1. Singh, A. (2008). Essentials of physical education. Kalyani Publishers.
2. Kamlesh, M. L. (2006). Psychology in physical education and sport (3rd ed.). Metropolitan Book Co.
3. Mangal, S. K. (2009). *Psychology of sports performance*. Sports Publication.

E-resources:

1. **https://www.who.int/health-topics/physical-activity**

	CO Description	PO	PSO
CO1	Understand and explain the importance of physical activity for mental and physical health.	---	---
CO2	Apply basic principles of exercise science in the routine life.	PO1(3)	PSO1(1)
CO3	Develop teamwork, discipline, and leadership through sports and group activities and collaborate effectively.	PO8(3)	PSO3(2)
CO4	Demonstrate independent learning in health, nutrition, and fitness-related topics.	PO11(2)	PSO3(2)

Semester II

MA25C02	Linear Algebra	L	T	P	C
		3	1	0	4

Course Objectives:

- To impart foundational knowledge in linear algebra essential for analysing and solving problems in engineering applications.
- To provide the knowledge on computation using software and interpret key linear algebra concepts using software.

Vector Spaces Introduction to Vector Spaces, Examples, Subspaces, Linear Combinations, Span, Generating Sets, Linear Dependence and Independence, Basis and Dimension, Dimension of Subspaces.

Activities: Open-Source software, exercises to test linear dependence and independence using rank, compute span and basis of a set of vectors, determine the dimension of subspaces, and illustrate the concept of subspace and basis in $\boldsymbol{R^2}/\boldsymbol{R^3}$ with visualization.

Linear Transformations and Diagonalization: Null space, Range, Dimension Theorem (statement only), Matrix representation of a linear transformation, Eigenvalues & Eigenvectors, Diagonalizability.

Activities: Open-Source software, exercises to compute the matrix representation of a linear transformation, find the null space and range of a matrix, and compute eigenvalues and eigenvectors of a matrix.

Inner Product Spaces: Inner product, Norms, Cauchy, Schwarz inequality, Gram, Schmidt orthogonalization, Simple problems (up to $\boldsymbol{R^3}$).

Activities: Open-Source software, exercises to compute inner products and vector norms.

Matrix Decomposition: Orthogonal transformation of a symmetric matrix to diagonal form - Positive definite matrices, QR decomposition, Singular Value Decomposition (SVD), Least squares solutions- simple problems (up to $3 \times 3\ matrices$).

Activities: Open-Source software, exercises to check if a matrix is positive definite, perform QR decomposition and SVD using built-in functions.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%.

Assessment Methodology: Assignment (20%), Software activity (20%), Quiz (20%), Internal Examinations (40%).

References:

1. Friedberg, S. H., Insel, A. J., & Spence, L. E. (2022). *Linear algebra*. Pearson.
2. Lay, D. C., Lay, S. R., & McDonald, J. J. (2020). *Linear algebra and its applications with MATLAB*. Pearson.
3. Bronson, R. (2011). *Schaum's outline of matrix operations*. McGraw-Hill Education.
4. Strang, G., & Thomson, R. (2005). *Linear algebra and its applications*. Brooks/Cole.
5. Lipschutz, S., & Lipson, M. (2009). *Schaum's outline of linear algebra*. McGraw-Hill.

Kreyszig, E. (2018). *Advanced engineering mathematics*. Wiley India.

	Description of CO	PO	PSO
CO1	Explain the fundamental concepts of Linear Algebra.	---	
CO2	Compute and interpret eigenvalues and eigenvectors.	PO1(3)	PSO1(2)
CO3	Apply inner product concepts and perform orthogonalization.	PO1 (3)	PSO1(1)
CO4	Compute least squares solutions of linear system of equations.	PO1 (2) PO2 (2)	PSO3(1)
CO5	Use MATLAB to implement and validate key linear algebra concepts	PO5 (1) PO11 (1)	PSO2(2)

ME25C02	Engineering Mechanics	L	T	P	C
		3	1	0	4

Course Objectives:

- To introduce the fundamental concepts and principles of statics related to forces acting on particles and rigid bodies.
- To develop the ability to formulate and apply equilibrium equations for particles and rigid bodies in two and three dimensions.
- To enable students to analyse force systems through vector resolution and calculation of moments and couples.

Statics of Particles: Resultant of forces in a plane, Equilibrium of a particle in a plane, Addition of concurrent forces in space, Equilibrium of a particle in space.

Activities: Assignments and Quiz on resultant forces, Solving of GATE questions.

Statics of Rigid Bodies: Concept of Free Body Diagram, Equivalent systems of forces, Transmissibility, Moment of a force about a point and an axis, Couples and force-couple systems, Equilibrium of rigid bodies in two and three dimensions, Principle of virtual work.

Activities: Virtual demonstration of rigid bodies, Solving of GATE questions.

Moments of Inertia: First moments of areas and lines, Centroids of composite areas and lines, Theorems of Pappus-Guldinus, Second moment of area, Parallel axis theorem, Rectangular and Polar Moments of inertia of composite areas, Radius of Gyration, Product of Inertia, Principal Axes and Principal Moments of Inertia, Mass moments of inertia of thin plates.

Activities: Virtual Simulation of Moment of Inertia, Principal Axes Determination, Solving of GATE questions.

Friction: Laws of friction, Coefficients of Friction, Angles of Friction, Types of Friction Problems, Wedges and Ladder friction, Belt friction.

Activities: Virtual Demonstration of Friction in belts and pulleys, Solving of GATE questions

Weightage:

Continuous Assessment: 40%
End Semester Examinations: 60%

Assessment Methodology:

Quiz - 10%, Assignments - 20%, Solving of GATE questions (20%) and Internal Examinations - 50%

References:

1. Beer, F. P., Johnston Jr., E. R., DeWolf, J. T., & Mazurek, D. F. (2015). Mechanics of Materials. McGraw-Hill Education.
2. Meriam, J. L., & Kraige, L. G. (2018). *Engineering Mechanics: Statics and Dynamics*. Wiley.

3. Pytel, A., & Kiusalaas, J. (2014). Engineering Mechanics (Indian Edition). Cengage Learning India.

E-resources:

1. Moment of Inertia Calculator – https://skyciv.com/free-moment-of-inertia-calculator/
2. OpenStax – University Physics Volume 1 – https://openstax.org/books/university-physics-volume-1/pages/10-4-moment-of-inertia-and-rotational-kinetic-energy
3. Engineering Mechanics, Dr. Dwarakish. G. S. – https://onlinecourses.swayam2.ac.in/ntr24_ed75/preview

	CO Description	PO	PSO
CO1	Explain the principles of statics in determination of forces acting on particles and rigid bodies.	---	---
CO2	Apply equilibrium conditions to predict the behaviour of particles and rigid bodies under various force configurations	PO1(3)	PSO1(3)
CO3	Analyse various systems through resolution of forces and moments.	PO2(2)	PSO1(2)
CO4	Demonstrate the ability to engage in adapting new techniques in the analysis of force and moments in a system.	PO11(1)	PSO2(2) PSO3(1)

EE25C01	Basic Electrical and Electronics Engineering	L	T	P	C
		3	0	0	3

Course Objectives:

- To impart foundational knowledge in principles and applications of electrical and electronics engineering.

DC Fundamentals: Current and Voltage sources, Resistance, Inductance and Capacitance; Ohm's law, Kirchhoff's law, Series parallel combination of R, L and C components, Voltage Divider and Current Divider Rules.

Activities: Virtual Demonstration of electrical laws & circuits, Hands-on Breadboarding, Solving GATE questions.

AC Fundamentals: Faraday's Laws of Electro-magnetic Induction, Definition of Self and Mutual Inductances, Generation of sinusoidal voltage, Instantaneous & RMS values of sinusoidal signals, Introduction to 3-phase systems, Electrical Safety, Fuses and Earthing.

Activities: Virtual Demonstration of electromagnetic induction, Measurement of instantaneous and RMS values of AC signals, Solving GATE questions.

Electric Machines: DC Machines, Transformers, Star and delta Connections, Three phase Induction motors, Synchronous Generators, Single Phase Induction Motors, Stepper Motor, Universal Motor and BLDC motor.

Activities: Virtual demonstration of step-up and step-down transformers, Virtual working models of Universal and BLDC motors, Solving GATE questions.

Semiconductor Devices: PN junction diodes, Zener Diode, Voltage regulator, BJT & FET Transistors, Timers, Operational Amplifiers.

Activities: Virtual demonstration of V-I characteristics of PN junction and Zener diodes using simulation, inverting/non-inverting amplifiers, Solving GATE questions.

Digital Electronics: Boolean algebra, Basic and Universal Gates, adders, multiplexers, demultiplexers and flip-flops.

Activity: Online logic gate simulators, Solving GATE questions.

Microcontrollers: Introduction, Architecture, Potential Applications.

Activities: Physical demonstration of a microcontroller and online simulation of microcontroller.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (5%), Assignments (25%), GATE Questions (20%), Internal Examinations (50%)

E-resources:

1. https://archive.nptel.ac.in/courses/108/106/108106172/
2. Circuit Simulator – https://www.falstad.com/circuit/

	Description of CO	PO	PSO
CO1	Understand and explain basic electrical and electronic concepts.	---	
CO2	Apply and analyse electrical circuits in real-time applications.	PO1 (3) PO2 (1)	PSO1(2)
CO3	Identify and utilise key electronic devices used in engineering applications	PO2 (2)	PSO1(2)

PH25C05	Applied Physics (ME) – II	L	T	P	C
		2	1	0	3

Course Objective(s):

- To impart fundamental knowledge of rigid body dynamics, thermal physics, phase transitions, and functional materials.
- To provide analytical abilities for evaluating physical phenomena in mechanical engineering applications.

Rigid Body Dynamics: Centre of mass – Moment of inertia (circular disc, solid cylinder, hollow cylinder, solid sphere, hollow sphere), Gear, shaft, gyroscope

Activities: Demonstration of moment of inertia of Gear, shafts and Gyroscopes.

Thermal Physics: Thermal conductivity –Transient plane source method, Transient Line Source method- Forbe's method - conduction through compound media, Laws of Thermodynamics .

Activities: Demonstration of thermal conductivity of insulators

Phase Transitions: Solid solutions - single component system, binary phase diagrams - iron-carbon equilibrium diagram, T-T-T-diagram - heat treatment of steels – hardening techniques

Activities: Demonstration of Hardening of steels

Functional Materials: Ceramics – Composites, Fiber Reinforced Plastics, Metallic Glasses, LED Characteristics

Activities: Demonstration of LED working and its characteristics.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (10%), Assignments (30%), Flipped Class (10%), Internal Examinations (50%)

References:

1. Mathur, D. S. (2008). *Elements of properties of matter*. S. Chand.
2. Brij Lal, & Subramaniyan, N. (2018). *Heat, thermodynamics and statistical physics*. S. Chand.
3. Raghavan, V. (2009). *Physical metallurgy: Principles and practice*. PHI Learning.
4. Askeland, D. (2010). *Materials science and engineering*. Brooks/Cole.

E-resources:

1. Moment of Inertia: https://youtu.be/fDJeVR0o__w
2. Conduction: http://kcl.digimat.in/nptel/courses/video/112106155/L32.html
3. Iron –Carbon phase diagram - https://archive.nptel.ac.in/courses/113/104/113104068/
4. Gyroscope_ https://www.youtube.com/watch?v=FydJu1A1oeM gyroscope-131127011945-phpapp02.pdf
5. Hardening Techniques_ https://www.youtube.com/watch?v=ckCN9jGsdUY
6. LED Characteristics - https://youtu.be/IgapvczVyXs_

	Description of CO	PO	PSO
CO1	Explain the concepts of physics in mechanical engineering stream.	---	
CO2	Apply appropriate techniques in physics to solve engineering problems.	PO1(3)	PSO1(3)
CO3	Analyse physical systems and interpret data from the virtual studies in the core branches in mechanical engineering.	PO2(2)	PSO1(2)

CY25C03	Applied Chemistry (ME) – II	L	T	P	C
		2	0	0	2

Course Objective(s):

- To impart knowledge and expose to applications of chemistry in mechanical engineering stream.
- To explore the mechanisms and working principles of smart materials and coatings with real-world applications.

Functional Materials: Types, Smart coatings, Mechanisms, Sustainable energy materials.

Activities: Seminar on recent development in functional materials (e.g., smart coatings, self-cleaning surfaces), Infographic Design of functional nanomaterials.

Fuels: Classification, Chemical Composition, natural resources, Calorific Value - Alternative Fuels - Natural gas benefits.

Activities: Comparison of efficiency and emissions in fuels.

Composites: Matrix materials – Reinforcements, Hybrid composites, Engineering applications.

Activities: Design of a simple composite structure for a real engineering application (e.g., lightweight bike frame).

Lubricants: Types, Functions. Key properties, Synthetic lubricants, Mechanisms, Emerging lubricants.

Activities: Collection of lubricants used in real-world engineering systems (e.g., gears, engines, bearings), Virtual demonstration of lubricant viscosity testing.

Combustion: Reaction Kinetics, Stoichiometric combustion and air-fuel ratio calculations, Knocking and Anti-knocking agents, Hydrogen combustion, Flue Gas analysis.

Activities: Virtual simulation of flue gas analysis and gas composition, Calculation of air-fuel ratio.

Adhesives: Adhesion Mechanisms, Classification, Bond strength, Industrial adhesives.

Activities: Adhesion of thermal pads on different Integrated circuits, Industrial adhesives.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (5%), Seminar (5%), Assignments (30%), Flipped Class (10%), Internal Examinations (50%)

References:

1. Palanna, O. G. (2009). *Engineering chemistry*. McGraw-Hill Education (India) Pvt. Ltd.
2. Cheong, K. Y., Impellizzeri, G., & Fraga, M. A. (2018). *Emerging materials for energy conversion and storage*. Elsevier.
3. Jain, P. C., & Jain, M. (2013). *Engineering chemistry*. Dhanpat Rai Publishing Company (P) Ltd.

Sharma, S. C. (2000). *Composite materials*. Narosa Publishing House

	Description of CO	**PO**	**PSO**
CO1	Explain the major concepts of chemistry with regard to applications in mechanical systems.	---	
CO2	Apply the chemistry principles and evaluate the engineering materials in mechanical systems.	PO1(3)	PSO1(2)
CO3	Analyse and evaluate the performance and efficiency of mechanical systems.	PO2(3) PO3(1)	PSO2(2) PSO3(1)
CO4	Propose innovative solutions for real-world applications and challenges.	PO9(1)	

UC25H02	தமிழர்களும் தொழில்நுட்பமும் / Tamils and Technology	L	T	P	C
		1	0	0	1

நெசவு மற்றும் பானைத் தொழில்நுட்பம்: சங்க காலத்தில் நெசவுத் தொழில், பானைத் தொழில்நுட்பம், கருப்பு சிவப்பு பாண்டங்கள், பாண்டங்களில் கீறல் குறியீடுகள்.

வடிவமைப்பு மற்றும் கட்டிடத் தொழில்நுட்பம்: சங்க காலத்தில் வடிவமைப்பு மற்றும் கட்டுமானங்கள் & சங்க காலத்தில் வீட்டுப் பொருட்களில் வடிவமைப்பு, சங்க காலத்தில் கட்டுமான பொருட்களும் நடுகல்லும் – சிலப்பதிகாரத்தில் மேடை அமைப்பு பற்றிய விவரங்கள், மாமல்லபுரச் சிற்பங்களும், கோவில்களும், சோழர் காலத்துப் பெருங்கோயில்கள் மற்றும் பிற வழிபாட்டுத் தலங்கள் – நாயக்கர் காலக் கோயில்கள், மாதிரி கட்டமைப்புகள் பற்றி அறிதல், மதுரை மீனாட்சி அம்மன் ஆலயம் மற்றும் திருமலை நாயக்கர் மஹால் – செட்டிநாட்டு வீடுகள், பிரிட்டிஷ் காலத்தில் சென்னையில் இந்தோ, சாரோசெனிக் கட்டிடக் கலை.

உற்பத்தித் தொழில் நுட்பம்: கப்பல் கட்டும் கலை, உலோகவியல், இரும்புத் தொழிற்சாலை, இரும்பை உருக்குதல், எஃகு, வரலாற்றுச் சான்றுகளாக செம்பு மற்றும் தங்க நாணயங்கள், நாணயங்கள் அச்சடித்தல், மணி உருவாக்கும் தொழிற்சாலைகள், கல்மணிகள், கண்ணாடி மணிகள், சுடுமண் மணிகள், சங்கு மணிகள், எலும்புத்துண்டுகள், தொல்லியல் சான்றுகள், சிலப்பதிகாரத்தில் மணிகளின் வகைகள்.

வேளாண்மை மற்றும் நீர்ப்பாசனத் தொழில் நுட்பம்: அணை, ஏரி, குளங்கள், மதகு, சோழர்காலக் குமுழித் தூம்பின் முக்கியத்துவம், கால்நடை பராமரிப்பு, கால்நடைகளுக்காக வடிவமைக்கப்பட்ட கிணறுகள், வேளாண்மை மற்றும் வேளாண்மைச் சார்ந்த செயல்பாடுகள், கடல்சார் அறிவு, மீன்வளம், முத்து மற்றும் முத்துக்குளித்தல், பெருங்கடல் குறித்த பண்டைய அறிவு, அறிவுசார் சமூகம்.

அறிவியல் தமிழ் மற்றும் கணித்தமிழ்: அறிவியல் தமிழின் வளர்ச்சி, கணித்தமிழ் வளர்ச்சி, தமிழ் நூல்களை மின்பதிப்பு செய்தல், தமிழ் மென்பொருட்கள் உருவாக்கம், தமிழ் இணையக் கல்விக்கழகம், தமிழ் மின் நூலகம், இணையத்தில் தமிழ் அகராதிகள், சொற்குவைத் திட்டம்.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (20%), Assignments (30%), Internal Examinations (50%)

Text-Cum-Reference Books

1. தமிழக வரலாறு, மக்களும் பண்பாடும், கே.கே. பிள்ளை (வெளியீடு: தமிழ்நாடு பாடநூல் மற்றும் கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ் – முனைவர் இல. சுந்தரம். (விகடன் பிரசுரம்).
3. கீழடி – வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. பொருநை, ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils – The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.)
9. Keeladi – 'Sangam City C ivilization on the banks of river Vaigai' (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by: The Author)
11. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.

UC25H02	Tamils and Technology	L	T	P	C
		1	0	0	1

Weaving and Ceramic Technology: Weaving Industry during Sangam Age, Ceramic technology, Black and Red Ware Potteries (BRW), Graffiti on Potteries.

Design and Construction Technology: Designing and Structural construction House & Designs in household materials during Sangam Age, Building materials and Hero stones of Sangam age, Details of Stage Constructions in Silappathikaram, Sculptures and Temples of Mamallapuram, Great Temples of Cholas and other worship places,Temples of Nayaka Period, Type study (Madurai Meenakshi Temple), Thirumalai Nayaka rMahal, Chetti Nadu Houses, Indo, Saracenic architecture at Madras during British Period.

Manufacturing Technology: Art of Ship Building , Metallurgical studies, Iron industry, Iron smelting, steel, Copper and gold Coins as source of history - Minting of Coins, Beads making, industries Stonebeads, Glass beads, Terracotta beads, Shell beads / bone beats, Archeological evidences, Gem stone types described in Silappathikaram.

Agriculture and Irrigation Technology: Dam, Tank, ponds, Sluice, Significance of Kumizhi Thoompuof Chola Period, Animal Husbandry - Wells designed for cattle use , Agriculture and Agro Processing -Knowledge of Sea -Fisheries, Pearl, Conche diving, Ancient Knowledge of Ocean -Knowledge Specific Society.

Scientific Tamil & Tamil Computing: Development of Scientific Tamil, Tamil computing, Digitalization of Tamil Books, Development of Tamil Software, Tamil Virtual Academy, Tamil Digital Library, Online Tamil Dictionaries, Sorkuvai Project.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (20%), Assignments (30%), Internal Examinations (50%)

Text-Cum-Reference Books

1. தமிழக வரலாறு, மக்களும் பண்பாடும், கே.கே. பிள்ளை (வெளியீடு: தமிழ்நாடு பாடநூல் மற்றும் கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ், முனைவர் இல. சுந்தரம். (விகடன் பிரசுரம்).
3. கீழடி, வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. பொருநை, ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils, The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies.)
9. Keeladi , 'Sangam City C ivilization on the banks of river Vaigai' (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by: The Author)
11. Porunai Civilization (Jointly Published by: Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigai (R.Balakrishnan) (Published by: RMRL) – Reference Book.

ME25C05	Re-Engineering for Innovation	L	T	P	C
		0	0	4	2

Course Objectives:

- To cultivate foundational skills in prototyping, and automation for development of prototypes with real-world applications.
- To provide a comprehensive, hands-on exposure to product development through reverse engineering concepts.

Bootcamp 1: Introduction to Product Development, Reverse Engineering, Overview of the product lifecycle, Hands-on disassembly of simple products, Practice of basic measurements and sketching, Introduction to CAD modeling of disassembled parts, Virtual assembly of parts.

Bootcamp 2: Embedded System Programming (Open-source platforms), Practice of interfacing sensors, reading data, automation in home, healthcare and agriculture.

Reverse Engineering: Sketch and prototype alternative designs, Group brainstorming sessions, Manufacture prototype parts using 3D printing and / or workshop tools, Assemble prototype product.

Weightage: Continuous Assessment: 60%, End Semester Examinations: 40%

Assessment Methodology: Project (30%), Assignment (10%), Practical (30%), Internal Examinations (30%)

References:

1. Wang, W. (2010). Reverse engineering: Mechanisms, structures, systems & materials. CRC Press.
2. Margolis, M. (2020). Arduino cookbook: Recipes to begin, expand, and enhance your projects (3rd ed.). O'Reilly Media.

E-resources:

1. GrabCAD – https://grabcad.com/
2. GitHub – https://github.com/

	Description of CO	PO	PSO
CO1	Understand the product development lifecycle, including stages such as concept generation, design, prototyping, and testing.	---	
CO2	Apply reverse engineering techniques to analyze and document existing products.	PO1 (3) PO2 (2)	PSO1(2)
CO3	Collaborate in teams to fabricate prototypes using appropriate tools.	PO5 (2) PO8 (1) PO9 (1)	PSO3(3)

	Description of CO	PO	PSO
CO4	Engage in independent learning and continuously adapt to emerging technologies in product design	PO11(2)	PSO2(2) PSO3(2)

EN25C02	English Essentials – II	L	T	P	C
		1	0	2	2

Course Objectives:

- Enable learners to improve fluency and accuracy in spoken and written communication.
- Develop learners' ability to articulate ideas clearly and effectively in formal and informal spoken interactions.
- Help learners construct well-organised written documents relevant to academic and workplace contexts.

Oral Communication: Types (Verbal and Nonverbal), Interpersonal and group communication, Telephonic conversation.

Suggested Activities: Short presentations, Debates, Formal Speeches (Welcome, Vote of Thanks and introducing guests), Listen and respond to short podcasts.

Business Correspondence: Email Communication, Formal Letters (Types), Business Meeting.

Suggested Activities: Email and letter writing (Complaint, request, permission), Agenda, minutes of the meeting.

Academic Writing: Paraphrasing, Summarizing, Essay Writing, Instructions and Recommendations.

Suggested Activities: Essay writing (Cause and effect, argumentative, persuasive), User guides/ manuals, policy document.

Team Work: Leadership Skills (Team building, Team Leader, Team player), Negotiation and Problem solving skills

Suggested Activities: SWOT Analysis, Brainstorming and Group discussions.

Weightage: Continuous Assessment: 50%, End Semester Examinations: 50%

Assessment Methodology: Worksheets (10%), Group Activity (20%), Report Writing (20%), Internal Examinations (50%)

References:

1. Koneru Aruna. (2020*). English Language Skills for Engineers*. McGraw Hill Education.
2. Taylor, Shirley & Chandra .V. (2010). *Communication for Business A Practical Approach*. India: Pearson Longman.
3. Ian Badger, et al., (2014). *Listening: B2 (Collins English for Life: Skills)*, Collins.
4. Raymond Murphy (2019), *Grammar in Use*, Cambridge University Press.

E-Resources:

1. Communication for Business Success - https://open.umn.edu/opentextbooks/textbooks/8
2. TED Talks – https://www.ted.com/

	Description of CO	PO	PSO
CO1	Understand the importance of communication and drafting skills in engineering and technology.	---	
CO2	Apply listening strategies to comprehend spoken English in various contexts.	PO1(3)	PSO3(2)
CO3	Participate actively in group discussions by analysing critically from different views.	PO2(2) PO8(1)	PSO3(3)
CO4	Create written reports coherently for various purposes.	PO9(2)	PSO3(2)
CO5	Adapt communication styles to global, multicultural environments.	PO11(1)	PSO2(2)

UC25A03	Life Skills for Engineers – II	L	T	P	C
		1	0	2	1

Course Objectives:

- To impart and cultivate analytical reasoning, innovative thinking, effective collaboration, and ethical leadership to prepare students for complex challenges in professional and personal environments.

Critical Thinking: Creativity, Critical Thinking, Collaboration, Problem Solving, Decision Making, Imagination, Intuition, Experience, Sources of Creativity, Lateral Thinking, Myths of creativity, Critical thinking Vs Creative thinking, Convergent & Divergent Thinking, Critical reading & Multiple Intelligence.

Activities: Two-Brainstorm Method, “30 Circles” Challenge, “Desert Survival” Simulation, Lateral thinking riddles and puzzles, "What If?" Scenario Writing, Fast vs. Slow Thinking Game, Creativity Myth Busters

Problem Solving: Techniques, Six Thinking Hats, Mind Mapping, Forced Connections. Analytical Thinking, Numeric, symbolic, and graphic reasoning. Scientific temperament and Logical thinking.

Activities: Case study analysis, Escape Room challenge.

Leadership: Leadership Styles & Self-Assessment, Communication & Active Listening, Decision-Making & Responsibility, Teamwork & Delegation, Empathy, Integrity & Conflict Management, Vision, Motivation & Goal-Setting.

Activities: Crisis Leadership Simulation, Tower Challenge, Leadership Dilemmas Role-Play, Team Vision Board

Weightage: Continuous Assessment: 100%

Assessment Methodology: Assignments (20%), Flipped Class & Worksheets (10%), Practical (30%), Internal Examinations (40%)

References:

1. De Bono, E. (2017). *Six thinking hats*, Little, Brown Book Group.
2. Facione, P. A. (2015). *Critical thinking: What it is and why it counts*. Insight Assessment.
3. Kahneman, D. (2011). *Thinking, fast and slow*. Farrar, Straus and Giroux.
4. Whetten, D. A., & Cameron, K. S. (2016). *Developing management skills*. Pearson.

	Description of CO	PO	PSO
CO1	Explain the importance of leadership and management skills in life.	---	
CO2	Apply and demonstrate creative thinking techniques to generate innovative solutions.	PO7 (3)	PSO1(1) PSO2(1)
CO3	Exhibit effective collaboration and communication skills through teamwork, active listening, and conflict resolution strategies.	PO8 (2)	PSO3(3)
CO4	Integrate scientific temperament and logical reasoning into c problem solving in engineering and real-world contexts.	PO11 (2)	PSO2(1) PSO3(2)

UC25A04	Physical Education - II	L	T	P	C
		0	0	4	1

Course Objectives:

- To impart knowledge on gymnastic exercises and pressing needs for upskilling in a particular game.

Basic gymnastics exercises: Warming up, Suitable exercise, Lead up games, Safety education, Movement education, Balanced Walk, execution, floor exercise, tumbling/acrobatics, grip, release, swinging, parallel bar exercise, horizontal bar exercise, flic-flac-walk and pyramids.

Upskilling in any one of the athletics: Broad Jump, High Jump, Triple Jump, Relay Sprints, Javelin Throw, Discuss Throw, Shot Put, Short and Long-distance Running.

Advance skills in any one of the indoor/outdoor games, which has been opted by the student in the I semester.

Weightage: Continuous Assessment: 100%

Assessment Methodology: Attendance (60%), Quiz (10%), Participation in Sports and Games (20%) and Viva Voce (10%)

References:

1. Singh, A. (2008). Essentials of physical education. Kalyani Publishers.
2. Kamlesh, M. L. (2006). Psychology in physical education and sport (3rd ed.). Metropolitan Book Co.
3. Mangal, S. K. (2009). *Psychology of sports performance*. Sports Publication.
4. Kandappan, K. (2004). *Foundations of physical education*. Friends Publications.

E-resources:

1. **https://www.who.int/health-topics/physical-activity**

	CO Description	PO	PSO
CO1	Understand and explain the importance of physical activity for mental and physical health.	---	
CO2	Apply safety principles and methods during sports activities.	PO1(3)	PSO3(1)
CO3	Develop teamwork, discipline, and leadership through sports and group activities and collaborate effectively.	PO8 (3)	PSO3(2)
CO4	Demonstrate the advanced technical skills and strategic understanding in the game of their interest.	PO11(1)	PSO3(2)

Foreign Language^

UC25F01	Deutsch – I	L	T	P	C
		1	0	2	1

Course Objectives:

- To impart fundamentals of the Deutsch language, including reading, writing systems, pronunciation, and speaking.

Basics & Introduction: German alphabet and pronunciation, Basic greetings and farewells, Introducing yourself and others (Ich heiße…, Wer bist du?), Numbers 1–100 and days of the week, Personal pronouns (ich, du, er, sie…), Sentence structure (SVO word order).

Activities: Alphabet spelling game, short skits, Use color-coded cards for SVO sentences.

Grammar Essentials & Everyday Vocabulary: Present tense of regular verbs (spielen, arbeiten, machen…), Common irregular verbs: sein (to be), haben (to have), gehen, kommen, Articles and gender (der, die, das; ein, eine), Simple questions and negation (nicht, kein), Describing people and things: adjectives and colors, Family, school, food, and common objects vocabulary.

Activities: Conjugate regular and irregular verbs, “Question Chain” game, Create a simple family tree.

Everyday Communication in German: Asking for and giving directions, Telling the time and talking about schedules, Ordering food and drinks at a café or restaurant, Talking about hobbies, weather, and daily routines, Listening to short conversations and responding appropriately, Introduction to German culture and formal/informal language use (du vs Sie).

Activities: Ordering food and drinks, Give directions, Formal / Informal greetings, Do’s and Don’ts.

Weightage: Continuous Assessment: 100%

Assessment Methodology: Assignments (30%), Quiz (10%) and Internal Examinations 60%

References:

1. Funk, H., Kuhn, C., & Demme, S. (2015). Menschen A1: Deutsch als Fremdsprache Kursbuch. Hueber Verlag.

	CO Description	PO	PSO
CO1	Understand simple spoken Deutsch in everyday contexts.	---	
CO2	Communicate with widely used Deutsch words effectively.	PO9 (2)	PSO3(2)
CO3	Develop the skills necessary for self-directed learning and continuous improvement in Deutsch language.	PO11 (1)	PSO3(2)

UC25F02	Japanese – I	L	T	P	C
		1	0	2	1

Course Objectives:

- To impart fundamentals of the Japanese language, including reading, writing systems, pronunciation, and speaking.

Writing Systems & Basic Communication: Introduction to Hiragana: vowels, basic characters, reading & writing, Introduction to Katakana: basic characters and usage, Basic greetings and farewells (こんにちは, おはようございます, さようなら), Introducing yourself (名前、出身、年齢), Basic sentence structure: Subject–Object–Verb, Numbers 1–100, days of the week, classroom expressions.

Activities: Flashcard games and writing drills, Self-introduction, Numbers & date-matching, Greeting expressions, Listening to audio.

Grammar & Everyday Vocabulary: Particles: は (wa), を (wo), の (no), へ (e), に (ni), Present tense verbs: です, ます-form conjugation (たべます、のみます), Negative forms: ではありません, ません, Describing people and objects using adjectives (い and な), Question formation: なに、どこ、だれ、いつ, Vocabulary for family, food, colors, and basic actions.

Activities: Verb conjugation drills, Guessing game, Picture description, “Shopping” with food vocab and counters

Conversation & Cultural Etiquette: Talking about routines and schedules (daily verbs, time expressions), Asking and giving simple directions (〜はどこですか？), Ordering food and making polite requests (〜をください、〜をおねがいします), Expressing likes and dislikes (すき・きらい), Listening to short conversations and identifying key phrases, Introduction to formal/informal speech and Japanese etiquette.

Activities: Skits and role-plays, daily schedule, beginner-level dialogue, Group discussion on etiquette.

Activities: Practice worksheets and flashcards for hiragana, Writing drills and reading simple katakana words, Dialogue practice for greetings and self-introduction, Sentence construction exercises with basic SOV structure, Particle usage exercises and short dialogues, Role-play scheduling, shopping, and telling time, Verb conjugation drills for common verbs, Descriptive sentence exercises using adjectives, Practice Q&A dialogues forming questions and negations, Kanji writing practice and quizzes for basic characters, Vocabulary tests and conversational practice on daily topics, Oral presentations and listening comprehension quizzes.

Weightage: Continuous Assessment: 100%
Assessment Methodology: Assignments (30%), Quiz (10%) and Internal Examinations 60%
References: 1. Banno, E., Ikeda, Y., Ohno, Y., Shinagawa, C., & Tokashiki, K. (2011). Genki I: An integrated course in elementary Japanese. The Japan Times. 2. The Japan Foundation. (2017). Marugoto Japanese language and culture starter (A1) course book for communicative language activities. Goyal Publishers.

	CO Description	PO	PSO
CO1	Understand simple spoken Japanese in everyday contexts.	---	
CO2	Communicate with widely used Japanese words effectively.	PO9 (2)	PSO3(2)
CO3	Develop the skills necessary for self-directed learning and continuous improvement in Japanese language.	PO11 (1)	PSO3(2)

UC25F03	Korean – I	L	T	P	C
		1	0	2	1

Course Objectives:

- To impart fundamentals of the Korean language, including reading, writing systems, pronunciation, and speaking.

Fundamentals of Korean: Introduction to Hangul: consonants and vowels, Basic pronunciation and syllable formation, Common greetings and self-introductions, Numbers (Sino-Korean and Native Korean basics), Basic sentence structure (Subject-Object-Verb), Simple expressions (e.g., 감사합니다, 안녕하세요).

Activities: Writing and reading Hangul practice sheets, Pronunciation drills and audio repetition, Dialogue practice for greetings and self-introduction, Counting and number exercises.

Essential Grammar and Vocabulary: Particles (은/는, 이/가, 을/를) and usage, Basic verbs and present tense conjugation, Sentence patterns: affirmative, negative, interrogative, Common adjectives and descriptive sentences, Expressing possession and location, Asking simple questions (어디, 뭐, 누구).

Activities: Verb conjugation and sentence formation drills, Role-play conversations for shopping and daily routines, Descriptive writing and speaking exercises, Question and answer practice.

Everyday Korean Communication: Polite speech levels and honorifics introduction, Talking about time, dates, and schedules, Ordering food, shopping phrases, counting objects, Simple directions and transportation vocabulary, Listening practice with short dialogues, Cultural notes on etiquette and communication.

Activities: Role-play ordering at a restaurant or buying items, Listening comprehension exercises, Giving and asking for directions practice, Group conversations and presentations.

Weightage: Continuous Assessment: 100%

Assessment Methodology: Assignments (30%), Quiz (10%) and Internal Examinations 60%

References:

1. King, R., Yeon, J., & Brown, A. (2015). Elementary Korean (2nd ed.). Tuttle Publishing.
2. Cho, Y., Lee, H., Schulz, C., Sohn, H.-M., & Sohn, S.-O. (2001). Integrated Korean: Beginning 1. University of Hawai‘i Press.

	CO Description	PO	PSO
CO1	Understand simple spoken Korean in everyday contexts.	---	
CO2	Communicate with widely used Korean words effectively.	PO9 (2)	PSO3(2)
CO3	Develop the skills necessary for self-directed learning and continuous improvement in Korean language.	PO11 (1)	PSO3(2)

Semester III

MA25C09	Computational Differential Equations	L	T	P	C
		3	1	0	4

Course Objectives: The Objectives of the course are to equip students with the ability to formulate and solve various types of Ordinary and Partial Differential Equations (ODEs & PDEs) and to develop proficiency in applying numerical methods and computational tools for solving differential equations encountered in engineering problems.

First Order Ordinary Differential Equations: Basic concepts of ordinary differential equations–Formation of first order differential equations from physical problems–First order and first-degree ODE: Variables separable–Exact equations–Leibnitz's equation – Bernoulli's equation – Numerical Methods: Solving first order ODE by Euler's formula, Taylor series and Runge Kutta method of 4th order.

Activities: Application and Visualization of the first order ODE using open-source software and solving Competitive Examination questions

Higher Order Ordinary Differential Equations: Formation of second order differential equations from physical problems–Linear equations of second and higher order with constant coefficients – Euler's linear equations – Method of variation of parameters–Numerical Methods: Solving second order ODE by Runge-Kutta method of 4th order and finite difference method.

Activities: solving ordinary differential equations using open-source software and solving of Competitive Examination questions

First Order Partial Differential Equations: Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions - Solution of PDE by variable separable method – Solution of standard types of first order partial differential equations (excluding reducible to standard types) – Lagrange's linear equation – Numerical Methods for first order PDE by Method of lines with Runge-Kutta method of order fourth.

Activities: Application and Visualization of the Partial Differential Equations using open-source software and solving Competitive Examination questions

Higher Order Partial Differential Equations: Linear homogeneous partial differential equations of second and higher order with constant coefficients. Numerical Methods: Finite difference techniques for the solution of two-dimensional Laplace's and Poisson's equations on rectangular domain—Solution of one-dimensional heat equation using Bender Schmidt and Crank Nicholson difference schemes—Solution of one-dimensional wave equation by explicit scheme.

Activities: Application of the second order PDE using open-source software of 1-D wave, 1-D heat equations, Laplace and Poisson equations and Solving Competitive Examination questions

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%.

Assessment Methodology: Assignments (20%), Solution to application-oriented problems using software (20%), Solving Competitive Examination questions (20%), Internal Examinations (40%).

References:

1. Erwin Kreyszig, Advanced Engineering Mathematics, (11th ed.), John Wiley & Sons, USA,2025.
2. Joe D. Hoffman, Numerical *Methods for Engineers and Scientists* (3rd ed.). CRC Press (Taylor & Francis Group), 2001.
3. Steven C. Chapra and Raymond P. Canale. *Numerical Methods for Engineers* (8th ed.). McGraw-Hill Education, 2021.
4. William E. Boyce and Richard C. DiPrima. Elementary Differential Equations and Boundary Value Problems, (11thed.). John Wiley & Sons, USA,2017.
5. Grewal B.S., "Higher Engineering Mathematics", Khanna Publishers, New Delhi, 45th Edition, 2026.
6. Randall J. LeVeque, Finite Difference Methods for Ordinary and Partial Differential Equations, SIAM, 2007.

E-resources:

1. https://ocw.mit.edu/courses/18-03-differential-equations-spring-2010/
2. https://archive.nptel.ac.in/content/syllabus_pdf/111107063.pdf
3. https://github.com/patrickwalls/mathematicalpython

	Description of CO	PO	PSO
CO1	Explain first and higher-order ODEs using appropriate analytical and numerical techniques.	-	-
CO2	Solve higher-order PDEs using classical and finite difference methods.	PO1(3) PO5(3)	PSO1(2)
CO3	Apply various solution methods to differential equations arising in engineering contexts.	PO1 (3) PO5(3)	PSO2(1)
CO4	Interpretation of numerical solutions for partial differential equations and using open-source software.	PO2(3) PO5(3) PO11(2)	PSO2(1)

(1) = Low; (2) = Medium; (3) = High

ME25C07	Applied Engineering Mechanics	L	T	P	C
		3	0	0	3

Course Objectives:
To provide a comprehensive understanding of applied engineering mechanics principles for analyzing structures, machines, and mechanical systems. It develops the ability to solve engineering problems using analytical, energy, and momentum methods.

Analysis of Structures: Planar trusses, Assumptions, static determinacy, method of joints, method of sections. Introduction to 3D trusses. Plane Frames, analysis of member forces. Simple Machines, mechanical advantage, velocity ratio, efficiency.

Activities:
- Simulation of plane trusses using FEM softwares.
- Physical modeling of 2D trusses using wooden sticks
- Case Studies of Collapse of 3D Trusses

Method of Virtual Work: Principle of virtual work, analysis of planar machines/mechanisms, Mechanical efficiency of real machines, Potential energy and equilibrium, Stability of equilibrium of mechanical systems

Activities: Problem-solving assignments using computational tools/programming environments.

Kinetics of Particles: Energy and Momentum methods: Work Energy principle, Conservation of Energy principle, Impulse Momentum principle, Impacts – Direct and Oblique.

Activities:
Assignments using programming tools to calculate the motion parameters of particles and present the results in graphical form.

Plane Motion of Rigid Bodies: Energy and Momentum methods, Kinetic energy of rigid bodies, Linear and angular momentum principles, Eccentric impacts of rigid bodies

Activities:
- Problem-solving assignments using computational tools/programming environments
- Case studies of rigid body motion in engineering systems

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (5%), Assignments (25%), Review of GATE/ESE Questions (20%), Internal Examinations (50%)

References:

1. Ferdinand P. Beer, E. Russell Johnston, David Mazurek, Phillip J. Cornwell and Brian Self, Vector Mechanics for Engineers: Statics and Dynamics, 13th Edition, McGraw-Hill Education, 2025.
2. R. C. Hibbeler, Engineering Mechanics: Statics & Dynamics, 16th Edition,

Pearson Education, 2026.

3. James L. Meriam, L. G. Kraige and J. N. Bolton, Engineering Mechanics: Statics and Dynamics (An Indian Adaptation), 9th Edition, Wiley India, 2021.

E-resources:

1. https://nptel.ac.in/courses/112106180
2. https://nptel.ac.in/courses/112106286

CO	CO Description	PO	PSO
CO1	Explain fundamentals of statics, trusses, frames, simple machines, virtual work, and energy–momentum principles.	—	—
CO2	Apply statics, virtual work, and energy methods to determine forces, reactions, efficiency, and motion parameters.	PO1 (2), PO2 (2)	PSO1 (2), PSO2 (2)
CO3	Analyze trusses, frames, particles, and rigid bodies using force, energy, and momentum methods to evaluate system behavior.	PO1 (2), PO2 (2), PO4 (2)	PSO1 (2), PSO2 (2)
CO4	Evaluate stability, efficiency, and performance of mechanical systems and develop engineering solutions.	PO2 (2), PO3 (2), PO4 (2)	PSO1 (2), PSO2 (2)

(1) = Low; (2) = Medium; (3) = High

ME25301	Engineering Thermodynamics	L	T	P	C
		4	0	0	4

Course Objectives:

This course introduces the fundamental principles and laws of thermodynamics related to energy, heat, work, and properties of pure substances. It develops the ability to analyze thermodynamic systems using the First and Second Laws, and to evaluate system performance using entropy, exergy, and property relations. The course also emphasizes solving engineering problems through analytical approaches.

Basic Concepts: Continuum and macroscopic approach; thermodynamic systems (closed and open); thermodynamic properties and equilibrium; state of a system, state postulate for simple compressible substances, state diagrams, paths and processes on state diagrams, ideal gas equation of state; concept of temperature, zeroth law of thermodynamics, thermodynamic temperature scale.

Activities:

- Identify and report closed and open thermodynamic systems from daily life.
- Fabricate a transparent plunger–cylinder setup to demonstrate an adiabatic process.
- Demonstrate Boyle's law and Charles's law.

First Law of Thermodynamics: Concept of energy and its various forms, concepts of heat, work and different modes of work; reversible and irreversible processes, concept of moving boundary work; first law applied to elementary processes, closed systems and control volumes, steady and unsteady flow analysis.

Activities:

- Develop a program to compute moving boundary work for different polytropic processes and plot P–V diagrams.

Second Law of Thermodynamics: Limitations of the first law of thermodynamics, concepts of heat engines and heat pumps/refrigerators, Kelvin-Planck and Clausius statements and their equivalence; perpetual motion machines, Carnot cycle and Carnot principles/theorems; Clausius inequality and concept of entropy; microscopic interpretation of entropy, the principle of increase of entropy, T-s diagrams; second law analysis of control volume; exergy analysis of closed and open systems; basics of third law of thermodynamics.

Activities:

- Develop a program to generate P–v and T–s diagrams for a Carnot cycle and evaluate heat transfer, work, and efficiency.
- Develop a program to analyze energy balance, exergy balance, exergy destruction, and second-law efficiency of steady-flow systems.

Properties of Pure Substances: Pure substances and their phases; Phase-change processes of pure substances, property diagrams for phase-change processes; Gibbs phase rule, thermodynamic property tables and charts; real gases and van der Waals equation of state; compressibility factor, principle of corresponding states, generalized compressibility chart.

Activities:

- Develop a program to generate T–v, P–v, and P–T diagrams for water using thermodynamic property correlations.
- Develop a program to evaluate the ideal-gas approximation error for water vapour under varying conditions.

Thermodynamic Relations: Reciprocity and cyclic relation, Helmholtz and Gibbs functions, Maxwell relations; Clapeyron and Clapeyron–Clausius equations; volume expansivity, isothermal and adiabatic compressibility, Mayer relation; Joule- Thomson coefficient.

Activities:

Develop a program to numerically verify a Maxwell relation using thermodynamic property data.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (5%), Assignments (25%), Review of GATE/ESE Questions, Internal Examinations (50%)

References:

1. Cengel. Y. A. and Boles. M. A., “Thermodynamics: An Engineering Approach”, 10th Edition, McGraw-Hill, 2025.
2. Moran. M. J., Shapiro. H. N., Boettner. D. D. and Bailey. M. B., “Fundamentals of Engineering Thermodynamics”, 9th Edition, Wiley, 2018.
3. Nag. P. K., “Engineering Thermodynamics”, 7th Edition, McGraw-Hill Education, 2025.

E-resources:

1. https://nptel.ac.in/courses/112105220
2. https://nptel.ac.in/courses/112105123
3. https://nptel.ac.in/courses/112106310
4. https://nptel.ac.in/courses/112106320

CO	CO Description	PO	PSO
CO1	Explain fundamentals of thermodynamics including laws, properties of pure substances, and thermodynamic relations.	—	—
CO2	Apply thermodynamic principles to calculate heat, work, energy interactions, efficiencies, and properties.	PO1 (2), PO2 (2)	PSO1 (2), PSO2 (2)
CO3	Analyze thermodynamic systems using energy, entropy, and property relations to determine behavior and performance.	PO1 (2), PO2 (2), PO4 (2)	PSO1 (3), PSO2 (2)
CO4	Evaluate performance, efficiency, and irreversibility of thermodynamic systems and develop engineering solutions.	PO2 (2), PO3 (2), PO4 (2)	PSO1 (2), PSO2 (2)

(1) = Low; (2) = Medium; (3) = High

CE25C11	Strength of Materials	L	T	P	C
		3	0	2	4

Course Objective:

To understand the behaviour of materials under different loading conditions and analyze stresses, strains, bending, torsion, deflection, columns, and pressure vessels.

Stress, Strain and Deformation of Solids: Rigid bodies and deformable solids, Tension, Compression and Shear Stresses, Deformation of simple and compound bars, Thermal stresses, Elastic constants, Volumetric strains, Stresses on inclined planes, Principal stresses and principal planes, Mohr's circle of stress.

Activity: Assignments and Quiz problems on stress, strain and deformation of solids.

Practical

1. Tensile test on mild steel rod.
2. Hardness test on metals (Rockwell and Brinell Hardness Tests)

Transverse Loading on Beams and Stresses In Beam: Beams, Types, Transverse loading on beams, Shear force and Bending moment in beams, Cantilever, Simply-supported and over-hanging beams. Theory of simple bending Bending stress distribution, Load carrying capacity, Proportioning of sections, Flitched beams, Shear stress distribution.

Activity: Demonstration of practical applications of transverse loading on beams, Simulation of Shear Force and Bending Moment Diagrams using web-based analysis tools.

Practical

1. Double shear tests on metal rod
2. Impact test on metal specimen (Izod and Charpy)

Torsion: Theory of Torsion, Stresses and Deformations in Solid and Hollow Circular Shafts, Combined bending moment and torsion of shafts, Power transmitted to shaft, Shaft in series and parallel, Closed and Open Coiled helical springs – springs in series and parallel.

Activity: Assignments on torsion of solid and hollow circular shafts.

Practical: Torsion test on mild steel rod.

Deflection of Beams and Springs: Elastic curve, Governing differential equation, Double integration method, Macaulay's method, Area moment method, Conjugate beam method for computation of slope and deflection of determinant beams, Strain energy method for determinate beams, Maxwell's reciprocal theorem, Leaf springs.

Practical

1. Deflection test on metal beam. Verification of Maxwell's reciprocal theorem.
2. Deflection test on carriage spring.

Columns and Struts: Introduction, Classification of columns, Axially loaded compression members, Euler's crippling load theory, Derivation of Euler's critical load formulae for various end conditions, Equivalent length, Slenderness ratio, Euler's critical stress, Limitations of Euler's theory, Rankine, Gordon formula, Eccentric loading and Secant formula, Prof. Perry's formula.

Activity: Model making of light weight bridge.

Practical: Compression test on wood and brick.

Thin Cylinders, Spheres and Thick Cylinders: Stresses in thin cylindrical shell due to internal pressure, circumferential and longitudinal stresses, Deformation in thin cylinders, Spherical shells subjected to internal pressure, Deformation in spherical shells, Thick cylinders, Lame's theory.

Activity: Assignments and Quiz problems on Thin Cylinders, Sphere and Thick Cylinders.

Tasks:

T1. Design a crane beam used in construction to safely carry heavy loads, considering bending stress and shear force.

T2. Design a transmission shaft for an automobile to transmit required power without failure under torsion and combined loading.

T3. Design a vehicle suspension spring (helical spring) to absorb shocks and provide required comfort based on load and deflection criteria.

T4. Design a building column (steel or concrete) to safely support axial loads, considering buckling and stability conditions.

Weightage: Continuous Assessment: 50%, End Semester Examinations: 50%

Assessment Methodology:

Written Test (40%), Practical (30%), Activity (30%).

Practical: Lab Experiments (50%), Tasks (50%) (Each student must complete a minimum of two tasks).

Activity: Review of GATE/ESE Questions (10%), Mini Project/ Quiz/ Assignment Programs/ Flipped Class /Seminar Presentation (20%)

References:

1. R. K. Bansal, A Textbook of Strength of Materials, 7th Edition, Laxmi Publications, New Delhi, 2024.
2. R. K. Rajput, Strength of Materials: Mechanics of Solids, Revised Edition, S. Chand Publishing, New Delhi, 2022.
3. S. Ramamrutham and R. Narayanan, Strength of Materials, Revised Enlarged Edition, Dhanpat Rai Publishing Company, New Delhi, 2025.
4. R. Subramanian, Strength of Materials, 3rd Edition, Oxford University Press, New Delhi, 2017.
5. S. S. Rattan, Strength of Materials, 4th Edition, McGraw Hill Education, New Delhi, 2021.
6. B. S. Basavarajaiah and P. Mahadevappa, Strength of Materials in SI Units,

3rd Edition, Universities Press, New Delhi, 2010.
7. S. P. Timoshenko and D. H. Young, Elements of Strength of Materials, 5th Edition, East-West Press, New Delhi, 2017.

E-resources:

1. https://onlinecourses.nptel.ac.in/noc25_ce25/preview
2. https://nptel.ac.in/courses/112107147
3. https://archive.nptel.ac.in/courses/105/105/105105108/
4. https://ocw.mit.edu/courses/3-11-mechanics-of-materials-fall-1999/
5. https://ocw.mit.edu/courses/3-11-mechanics-of-materials-fall-1999/pages/lecture-notes/
6. https://www.vlab.co.in/ba-nptel-labs-mechanical-engineering
7. https://solidmechanics.org/
8. https://www.engineeringtoolbox.com/stress-strain-d_950.html
9. https://www.engineeringtoolbox.com/beam-stress-deflection-d_1312.html

CO	CO Description	PO	PSO
CO1	Describe fundamental concepts of stress, strain, deformation, bending, torsion, deflection, columns, and pressure vessels to demonstrate overall understanding of strength of materials.	—	—
CO2	Calculate stresses, strains, deformation, shear force, bending moment, and load-carrying capacity of structural members using relevant theories.	PO1 (2), PO2 (2)	PSO1 (2)
CO3	Analyze beams, shafts, springs, and columns under various loading conditions to determine stress distribution and structural response.	PO1 (2), PO2 (2), PO4 (2)	PSO1 (2), PSO2 (2)
CO4	Evaluate strength, stability, and performance of structural components and develop suitable solutions for engineering applications involving deformation and failure analysis.	PO2 (2), PO3 (2), PO5 (2)	PSO1 (2), PSO2 (2), PSO3 (1)

(1) = Low; (2) = Medium; (3) = High

ME25C08	Metallurgy and Materials Science	L	T	P	C
		3	0	0	3

Course Objective:

This course provides knowledge on phase diagrams, ferrous and non-ferrous materials, heat treatment processes, polymers, ceramics, composites, and mechanical behavior of engineering materials. It enables students to understand material selection, microstructure-property relationships, testing methods, and modern materials used in engineering applications.

Constitution of Alloys and Phase Diagrams: Constitution of alloys – Solid solutions, substitutional and interstitial – phase diagrams, Isomorphous, eutectic, eutectoid, peritectic, and peritectoid reactions, Iron – Iron carbide equilibrium diagram. Classification of steel and cast-Iron microstructure, properties and application.

Activity: Study of phase diagrams and microstructures.

Heat Treatment: Definition – Full annealing, stress relief, recrystallisation and spheroidising – normalizing, hardening and tempering of steel. Isothermal transformation diagrams – cooling curves superimposed on I.T. diagram – continuous cooling Transformation (CCT) diagram – Austempering, Martempering – Hardenability, Jominy end quench test -case hardening, carburizing, Nitriding, cyaniding, carbonitriding – Flame and Induction hardening – Vacuum and Plasma hardening – Thermo-mechanical treatments- elementary ideas on sintering.

Activity: Demonstration of heat treatment and hardenability test.

Ferrous and Non-Ferrous Metals: Effect of alloying additions on steel (Mn, Si, Cr, Mo, Ni, V,Ti& W) – stainless and tool steels – HSLA - Maraging steels – Grey, white, malleable, spheroidal – alloy cast irons, Copper and its alloys – Brass, Bronze and Cupronickel – Aluminium and its alloys; Al-Cu – precipitation strengthening treatment – Titanium alloys, Mg-alloys, Ni-based super alloys – shape memory alloys- Properties and Applications-overview of materials standards.

Activity: Alloy identification and material selection exercise.

Non-Metallic Materials: Polymers – types of polymers, commodity and engineering polymers – Properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA, ABS, PAI, PPO, PPS, PEEK, PTFE, Thermoset polymers – Urea and Phenol formaldehydes – Nylon, Engineering Ceramics – Properties and applications of Al2O3, SiC, Si3N4, PSZ and SIALON – intermetallics- Composites- Matrix and reinforcement Materials- applications of Composites - Nano composites**.**

Activity: Classification of polymers, ceramics, and composites.

Mechanical Properties and Deformation Mechanisms: Mechanisms of plastic deformation, slip and twinning – Types of fracture – fracture mechanics- Griffith's theory- Testing of materials under tension, compression and shear loads – Hardness tests (Brinell, Vickers and Rockwell), Micro and nano-hardness tests, Impact test lzod and charpy, fatigue and creep failure mechanisms.

Activity: Demonstration of tensile, hardness, and impact tests.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (5%), Assignments (25%), Review of GATE/ESE Questions, Internal Examinations (50%)

References:

1. Kenneth G.Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 9th edition ,2018.
2. Sydney H. Avner, Introduction to Physical Metallurgy, McGraw Hill Education, 2nd Edition, Reprint 2017.
3. Alavudeen N., Venkateshwaran N., and Winowlin Jappes J.T., A Textbook of Engineering Materials and Metallurgy, Laxmi Publications, Revised Edition, 2016.
4. Amandeep Singh Wadhwa and Harvinder Singh Dhaliwal, A Textbook of Engineering Material and Metallurgy, University Science Press, Revised Edition, 2018.
5. G.S. Upadhyay and Anish Upadhyay, Materials Science and Engineering, Viva Books Pvt. Ltd., 2020.
6. Raghavan V., Materials Science and Engineering, Prentice Hall of India Pvt. Ltd., 7th Edition, 2023.
7. William D. Callister Jr. and David G. Rethwisch, Materials Science and Engineering: An Introduction, Wiley India Pvt. Ltd., 10th Edition, 2020.

CO	CO Description	PO	PSO
CO1	Explain phase diagrams, heat treatment, engineering materials, and material properties.	—	—
CO2	Apply heat treatment methods, alloy selection, and material testing techniques.	PO1 (2), PO2 (2)	PSO1 (2)
CO3	Analyze microstructure, phase transformation, and failure behavior of materials.	PO1 (2), PO2 (2), PO4 (2)	PSO1 (2), PSO2 (2)
CO4	Evaluate the performance of metallic, ceramic, polymer, and composite materials.	PO2 (2), PO3 (2), PO5 (2)	PSO1 (2), PSO2 (2), PSO3 (1)

(1) = Low; (2) = Medium; (3) = High

EC25C17	Embedded Systems	L	T	P	C
		3	0	0	3

Course Objective:

This course introduces students to embedded system architecture, microcontrollers, and real-time operating systems. Students learn programming in C and Python, peripheral interfacing, and communication protocols to design and implement embedded applications for real-world automation systems.

Embedded Hardware Architecture: CISC Architecture: Introduction to MCS51 Family, 8051 Microcontroller – Architecture, Timers, Interrupts, Serial Data Communication.

RISC Architecture: Overview of PIC16F487x family, PIC16F877A – Architecture, Timers, Interrupts, Serial ports.

Activity: Simulate and compare basic programs (timers/interrupts) on 8051 and PIC microcontrollers.

Arm And Embedded Software Development Tools: Introduction to ARM – LPC4088 Architecture. Software Development Tools: IDE Tools, ISP Tools, ARM Development Tools.

Activity: Develop and debug a simple embedded application using an ARM-based IDE and development tools.

Wired Communication Interfaces: Wired Communication Protocols: Serial – RS232, RS485, I2C, SPI, USB; Parallel – IEEE 488.

Activity: Implement serial communication (UART/I2C/SPI) between a microcontroller and an external device.

Wireless Communication Interfaces: Wireless Communication Protocols: Bluetooth Classic, BLE, IEEE 802.15.4, Zigbee, IEEE 802.11, LoRaWAN.

Activity: Demonstrate wireless data transmission using a module (e.g., Bluetooth/Zigbee/Wi-Fi).

Real-Time Operating System: Operating System Basics: The Kernel and its subsystems, Kernel Space and User Space. Types and Functions of RTOS: Task, Process and Threads, Interrupt Handling, Multiprocessing and Multitasking, Task Scheduling. Comparative study of various RTOSs.

Activity: Simulate task scheduling and interrupt handling using an RTOS environment.

Embedded Programming and Peripheral Interfacing: Embedded C and Python Programming for Embedded Applications. Peripheral Interfacing: Input and Output Devices, ADC, DAC, PWM Generation, Sensor Interface.

Activity: Develop an embedded application interfacing sensors/actuators using open source tool.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (5%), Assignments (25%), Review of GATE/ESE Questions, Internal Examinations (50%)

References

1. Raj Kamal, Embedded Systems: Architecture, Programming and Design, 3rd Edition, McGraw Hill Education, 2017.
2. James K. Peckol, Embedded Systems: A Contemporary Design Tool, 2nd Edition, Wiley, 2019.
3. Tammy Noergaard, Embedded Systems Architecture: A Comprehensive Guide for Engineers and Programmers, 2nd Edition, Elsevier, 2012.
4. Han-Way Huang, Embedded System Design Using C8051, 1st Edition, Cengage Learning, 2009.
5. Rajib Mall, Real-Time Systems: Theory and Practice, 2nd Edition, Pearson Education, 2017.
6. Shibu K. V., Introduction to Embedded Systems, 3rd Edition, McGraw Hill Education, 2017.

E-Resources:

1. https://nptel.ac.in/courses/108105057
2. https://nptel.ac.in/courses/106105193
3. https://nptel.ac.in/courses/106105172
4. https://nptel.ac.in/courses/106105193
5. https://nptel.ac.in/courses/106105159

CO	CO Description	PO	PSO
CO1	Explain the architecture, components, programming, and communication protocols of embedded systems.	—	—
CO2	Apply embedded C/Python programming and peripheral interfacing techniques to implement embedded applications.	PO1 (2), PO5 (2)	PSO1 (2), PSO2 (2)
CO3	Analyze embedded systems for performance, task scheduling, and peripheral communication using real-time operating systems.	PO1 (2), PO3 (2), PO5 (2)	PSO1 (2), PSO2 (2)
CO4	Evaluate efficiency, reliability, and suitability of embedded systems, communication protocols, and RTOS-based solutions.	PO1 (2), PO2 (2), PO5 (2)	PSO1 (2), PSO2 (2), PSO3 (2)

(1) = Low; (2) = Medium; (3) = High

EN25C03	English Communication Skills Laboratory– I	L	T	P	C
		0	0	2	1

Course Objectives: The objectives of the course are to foster students' confidence and fluency in professional and social communication and to bridge the gap between academic English and industry expectations.

List of Activities

A. Elements of Effective Speaking and Listening

(i) Sharing life experience/ turning point in their life – SATORI

(ii) Situational Conversation – eg. Talking to a Senior about Internship Tips

(iii) Welcoming a Guest Speaker at a Seminar

(iv) Pictography to represent data using images or symbols

(v) B2-C1 Listening exercises include lectures, interviews, and discussions.

B. Mastering Presentations

(i) Presentation Skills – Non-verbal communication

(ii) Mini-Presentations: Topics like "My Dream Project," "Engineering in 2050," 3-minute technical pitches with logical flow

(iii) Technical Presentations with PPT

C. Group Discussion Strategies:

(i) Introduction to Group Discussions - Key skills for effective participation

(ii) Phases in a GD and Conversational Phrases in GD.

(iii) Group Discussions – Abstract and Factual topics

D. Resume & LinkedIn Optimization

(i) Building LinkedIn Profile – Drafting headlines and summaries

(ii) Social Media Optimisation

(iii) Preparing Video Resume

E. Podcast-Based Language Learning:

(i) Listening to podcast (motivational, career oriented, success stories)

(ii) Podcast Preparation – Purpose – Topic – Structure – Recording Tips - Publication of the Podcast

F. Mock Interviews and Communication Strategies:

(i) Listening – Job interview

(ii) Speaking – Mock interviews

Weightage: Continuous Assessment: 60%, End Semester Examinations: 40%.

Internal Assessment: 1. Listening (20 marks)
2. Video Resume (20 marks)
3. Creating a Podcast (30 marks)
4. Mock interview (30 marks)

End Semester Assessment: 1. Presentation with PPT (50 marks)
2. Group Discussion (50 marks)

References:

1. Floyd Kory, “Interpersonal Communication”, McGraw Hill Publication, 2023.
2. Bharadwaj Apoorva, “Leadership Communication Skills for Intercultural Management: Strategies for Effective Intercultural Management (Contemporary Themes in Business and Management)”, Routledge India; 1st edition, 2024.
3. Helen Spencer-Oatey and DomnaLazidou**, “**Making Working Relationships Work: The TRIPS Toolkit for Handling Relationship Challenges and Promoting Rapport”, Castledown Publishers,2023.
4. Presentations - Cambridge
5. Speaking Extra
6. Listening Extra – Miles Craven by Cambridge University Press
7. CVs, Resumes, and LinkedIn: A Guide to Professional English – Springer International Publishing

E-resources:

1. Train your mind to perform under pressure- Simon Sinek
https://curiosity.com/videos/simon-sinek-on-training-your-mind-to-perform-under-pressure-capture-your-flag/
2. Brilliant way one CEO rallied his team in the middle of layoffs
https://www.inc.com/video/simon-sinek-explains-why-you-should-put-people-before-numbers.html
3. Will Smith's Top Ten rules for success
https://www.youtube.com/watch?v=bBsT9omTeh0

	Description of CO	PO	PSO1
CO1	Communicate effectively in everyday professional situations with confidence		
CO2	Deliver well-organised and effective presentations	PO9(3)	PSO1(1) PSO3(2)
CO3	Participate in group discussions and express ideas clearly and confidently.	PO8(2) PO9(3)	PSO3(2)
CO4	Create professional video resumes and participate in interviews effectively.	PO9(2)	PSO3(3)
CO5	Create, record and publish motivational podcasts.	PO9(2) PO11(1)	PSO2(2) PSO3(3)

(1) = Low; (2) = Medium; (3) = High

SEMESTER–IV

CS25C16	Applied Data Science	L	T	P	C
		3	0	0	3

Course Objectives

This course introduces the core concepts of data science, including data preprocessing, exploratory analysis, and machine learning techniques. It aims to develop the ability to analyze and interpret data using modern tools for real-world problem solving while emphasizing ethical and responsible use of data.

Introduction and Mathematics for Data Science: Linear equations and solutions, Matrices and their Properties; Eigenvalues and eigenvectors; Matrix Factorizations, Inner products, Distance measures; Projections; Notion of hyperplanes; halfplanes. Descriptive statistics, Inferential statistics, Hypothesis testing, Regression analysis Statistical modelling.

Activity: Perform statistical analysis and regression on a dataset using programming language to understand data distribution and relationships.

Introduction and Mathematics for Data Science: Data Science Definition, Data Science lifecycle, Types of data (structured, unstructured, semi-structured), Data sources and data collection methods- Role of Data Scientist- Applications in healthcare, banking, e-commerce.

Data Preprocessing & Wrangling: Data cleaning (missing values, noise handling), Data transformation and normalization, Feature engineering, Data integration, Handling Outliers-Data pipelines-Tools: Python (Pandas, NumPy)

Activity: Clean and preprocess a real dataset by handling missing values, outliers, and applying normalization using Pandas.

Exploratory Data Analysis (EDA) & Visualisation: Descriptive statistics, Data visualisation techniques, Bar charts, histograms, box plots, Scatter plots, heatmaps, Correlation and covariance- Identifying patterns and trends, Tools: Matplotlib, Seaborn

Activity: Analyze and visualize a dataset using plots (histogram, scatter, heatmap) to identify patterns and correlations.

Machine Learning Techniques: Supervised Learning: Linear Regression, Logistic Regression, Decision Trees, k-NN, Unsupervised Learning: K-Means Clustering, Hierarchical Clustering, Model evaluation: Accuracy, Precision, Recall, F1-score, Confusion Matrix, Overfitting & Underfitting.

Activity: Build and evaluate a machine learning model (e.g., regression or classification) using Scikit-learn.

Applied Data Science & Case Studies: Real-world applications, Recommendation systems, Fraud detection, Traffic Route Optimisation, Customer segmentation, Introduction to Big Data concepts, Ethics in Data Science, Data

privacy and security basics.

Activity: Develop a mini project (e.g., house price prediction or customer segmentation) and present insights with ethical considerations.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%

Assessment Methodology: Quiz (5%), Assignments (25%), Review of GATE/ESE Questions (20%), Internal Examinations (50%)

References

1. Provost, F., & Fawcett, T. (2013). Data Science for Business: What You Need to Know About Data Mining and Data-Analytic Thinking. O'Reilly Media.
2. Wes McKinney, *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter*, 3rd Edition, O'Reilly Media, 2022.
3. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*, 3rd Edition, O'Reilly Media, 2022.
4. Joel Grus, Data Science from Scratch: First Principles with Python, 2nd Edition, O'Reilly Media, 2019.

E-resources:

1. https://onlinecourses.nptel.ac.in/noc25_cs120/preview
2. https://nptel.ac.in/courses/106106212
3. https://ocw.mit.edu/courses/6-0002-introduction-to-computational-thinking-and-data-science-fall-2016/
4. https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/
5. https://www.coursera.org/learn/what-is-datascience
6. https://www.coursera.org/specializations/data-science-python
7. https://pandas.pydata.org/docs/
8. https://numpy.org/doc/
9. https://scikit-learn.org/stable/user_guide.html

CO	CO Description	PO	PSO
CO1	Explain the concepts of data science lifecycle, mathematics, preprocessing, EDA, and ML techniques.	–	–
CO2	Describe data preprocessing, feature engineering, and visualization techniques.	PO1 (2), PO5 (2)	PSO2 (2)
CO3	Apply data preprocessing, visualization, and machine learning algorithms using tools.	PO2 (2), PO5 (2)	PSO2 (2)
CO4	Analyze datasets using statistical and analytical methods to derive insights.	PO2 (2), PO4 (2)	PSO2 (2), PSO3 (2)

(1) = Low; (2) = Medium; (3) = High

ME25C09	Kinematics and Dynamics of Machines	L	T	P	C
		3	0	2	4

Course Objectives:

The course introduces the fundamental concepts of mechanical linkages and mechanisms used in machines, including higher pairs such as cams and gear transmissions. It develops an understanding of force analysis in mechanisms, the effects of unbalanced forces, and the basic principles governing vibrations in mechanical systems.

Kinematic Analysis in Simple Mechanisms and Cams: Mechanisms, Terminology and definitions, kinematics inversions of 4 bar and slide crank chain, kinematics analysis in simple mechanisms, velocity and acceleration polygons.

Practical: Analysis of Four Bar Chain and Slider Crank Mechanisms,

Cams: Cam-terminology, different types of cams and followers, Cam design for different follower motion curves, graphical construction of cam profiles for different types of followers.

Practical: Cam displacement curves.

Gears and Geartrains: Gear terminology, law of toothed gearing, involute gearing, Gear tooth action, Interference and undercutting, geartrains, parallel axis gear trains, Epicyclic geartrains.

Practical: Experimental study of velocity ratios of simple, compound, Epicyclic and differential gear trains.

Static and Dynamic Force Analysis: Applied and Constrained Forces, Free body diagrams, Static equilibrium conditions, Static Force analysis in simple mechanisms, Dynamic Force Analysis in simple machine members, Inertia Forces and Inertia Torque, D'Alembert's principle.

Balancing of Rotating Masses and Gyroscope: Static and Dynamic balancing, Balancing of revolving masses, Balancing machines, Gyroscopes, Gyroscopic Couple, Gyroscopic stabilization, Gyroscopic effects in Automobiles, ships and airplanes.

Practical: Motorized gyroscope, Study of gyroscopic effect and couple.

Mechanical Vibrations: Vibrations-single degree of freedom system, equations of motion-undamped and damped free vibration of single degree of freedom system, logarithmic decrement, Forced vibration of single degree of freedom system, magnification factor, rotating unbalance-support harmonic excitation, critical speed of shafts

Practical:

- Undamped free vibrations of a single degree freedom spring-mass system.
- Torsional Vibration (Undamped) of single rotor shaft system
- Determination of critical speed of shafts.

Tasks:

T1. Design a four-bar mechanism for an application such as a windshield wiper or robotic arm to achieve a desired motion path.

T2. Design a cam and follower system for an IC engine valve mechanism to achieve specified motion (displacement, velocity, acceleration).

T3. Design a gear train system (simple or epicyclic) for an automobile gearbox or industrial machine to obtain required speed reduction and torque transmission.

T4. Analyze and design a rotating system (shaft with masses) for proper balancing and determine the critical speed to avoid excessive vibration in machines (e.g., turbine shaft or motor rotor).

Weightage: Continuous Assessment: 50%, End Semester Examinations: 50%.

Assessment Methodology:

Written Test (40%), Practical (30%), Activity (30%).

Practical: Lab Experiments (50%), Tasks (50%) (Each student must complete a minimum of two tasks).

Activity: Review of GATE/ESE Questions (10%), Mini Project/ Quiz/ Assignment Programs/ Flipped Class /Seminar Presentation (20%)

References:

1. S. S. Rattan, Theory of Machines, 5th Edition, McGraw Hill, 2019.
2. J. J. Uicker Jr., G. R. Pennock & J. E. Shigley, Theory of Machines and Mechanisms, 6th Edition, Cambridge University Press, 2023.
3. P. L. Ballaney, Theory of Machines and Mechanisms, 26th Edition, Khanna Publishers, 2024.
4. R. L. Norton, Kinematics and Dynamics of Machinery, 3rd Edition (SI Units), McGraw Hill Education, 2020.

E-resources:

1. https://nptel.ac.in/courses/112105268
2. https://nptel.ac.in/courses/112104114
3. https://nptel.ac.in/courses/112107212
4. https://nptel.ac.in/courses/112104121

CO	CO Description	PO	PSO
CO1	Understand basic mechanisms and solve kinematic analysis of various mechanisms	–	–
CO2	Evaluate static and dynamic forces in mechanical systems.	PO1 (2), PO2 (2)	PSO1 (2)
CO3	Analyze rotating masses for unbalance and examine the vibrational behavior of mechanical systems	PO2 (2), PO3 (2)	PSO1 (2)
CO4	Conduct experiments in groups and interpret the data	PO4 (2), PO8 (1)	PSO1 (2), PSO2 (2)

(1) = Low; (2) = Medium; (3) = High

CE25C12	Fluid Mechanics and Machinery	L	T	P	C
		3	0	2	4

Course Objective:

To introduce the students about properties of the fluids, behaviour of the fluids under statics, kinematics and dynamic conditions. To expose to the applications of the conservation laws to flow measurements, flow through pipes and forces on pipe bends. To impart basic knowledge to the students about the working principles and performance characteristics of hydraulic machineries.

Fluids Properties and Fluid Statics: Types and properties of fluids, Viscosity, Surface tension and Compressibility, Basics of Compressible flow Fluid statics, Concept of fluid pressure, Manometers, Forces on plane and curved surfaces, Buoyancy and floatation, Stability of floating bodies, metacentric height and its determination.

Activity: Mini case study on buoyancy applications in ships/submarines, Problem-solving using simulation tools for pressure distribution.

Practical: Determination of metacentric height

Fluid Kinematics and Dynamics: Kinematics: Classification of fluid flows, Eulerian and Lagrangian approach, concept of control volume and system, Velocity and acceleration, Stream function and velocity potentials, Flow nets; Dynamics: Application of control volume to continuity, energy and momentum, Euler's equation of motion along a stream line, Bernoulli's equation, Applications to velocity and discharge measurements, Linear momentum equation.

Activity: Group activity on Bernoulli-based engineering applications (aircraft wing, carburetor, atomizer, etc., software demonstration for flow patterns.

Practical: Determination of coefficient of discharge of a venturimeter.

Flow Through Pipes and Boundary Layer: Reynold's Experiment, Laminar flow through circular conduits, Hagen Poiseuille equation, Darcy-Weisbach equation, Moody diagram, Major and minor losses of flow in pipes, Hydraulic gradient and total energy gradient, Pipes in series and parallel, Equivalent pipes, Boundary layer concepts, types of boundary layer thickness.

Activity: Pipe network design exercise using campus/building water supply layout, Industrial case study on pipeline losses and leakage reduction.

Practical: Determination of friction factor for flow through pipes.

Dimensional Analysis: Fundamental dimensions, Dimensional homogeneity, Rayleigh's method and Buckingham Pi theorem, Dimensionless parameters, Similitude and model studies, Distorted and undistorted models

Activity: Model-making activity demonstrating similitude concepts, Group exercise on identifying dimensionless numbers in engineering systems.

Turbines: Impact of jets, Velocity triangles, Theory of rotodynamic machines, Classification of turbines, Pelton wheel, Francis turbine and Kaplan turbine, working principles, Work done by water on the runner, Efficiencies, Draft tube, Specific speed, Performance curves for turbines.

Activity: Virtual demonstration of Turbines.

Practical: Characteristics of Pelton wheel turbine

Pumps: Classification of pumps, Centrifugal pumps, working principle, Heads and efficiencies, Work done by the impeller, NPSH, Minimum speed to start the pump, Pumps connected in series and parallel, Performance curves, Reciprocating pump working principle, Indicator diagram and its variations, Air vessels - Work saved by air vessels.

Activity: Virtual demonstration of Pumps.

Practical:
1. Characteristics of centrifugal pumps
2. Characteristics of reciprocating pump

Tasks:

T1. Design a water supply pipeline system for a residential building, considering flow rate, head loss, and pipe sizing.

T2. Design a flow measurement system using a venturimeter or orifice meter for an industrial application to determine discharge accurately.

T3. Select and design a hydraulic turbine system (Pelton/Francis/Kaplan) for a small hydroelectric power plant based on site conditions.

T4. Design a pumping system (centrifugal or reciprocating) for irrigation or water transfer, considering head, discharge, and efficiency.

Weightage: Continuous Assessment: 50%, End Semester Examinations: 50%

Assessment Methodology:

Written Test (40%), Practical (30%), Activity (30%).

Practical: Lab Experiments (50%), Tasks (50%) (Each student must complete a minimum of two tasks).

Activity: Review of GATE/ESE Questions (10%), Mini Project/ Quiz/ Assignment Programs/ Flipped Class /Seminar Presentation (20%)

References:

1. P. N. Modi and S. M. Seth, Hydraulics and Fluid Mechanics including Hydraulic Machines, 23rd Edition, Standard Book House, New Delhi, 2022.
2. A. K. Jain, Fluid Mechanics including Hydraulic Machines, 12th Edition, Khanna Publishers, New Delhi, 2022.
3. S. K. Som, Gautam Biswas and S. Chakraborty, Introduction to Fluid Mechanics and Fluid Machines, 3rd Edition, McGraw Hill Education, Chennai, 2024.
4. B. S. Pani, Fluid Mechanics: A Concise Introduction, 2nd Edition, PHI Learning Private Limited, 2021.
5. A. K. Jain, Fluid Mechanics including Hydraulic Machines, 12th Edition, Khanna Publishers, New Delhi, 2022.
6. V. L. Streeter, E. B. Wylie and K. W. Bedford, Fluid Mechanics, 9th Edition, McGraw Hill Education, New Delhi, 2017.

E-resources:

1. https://onlinecourses.nptel.ac.in/noc25_ce107/preview
2. https://elearn.nptel.ac.in/shop/nptel/introduction-to-fluid-mechanics/
3. https://opencw.aprende.org/courses/mechanical-engineering/2-06-fluid-dynamics-spring-2013/
4. https://www.digimat.in/nptel/courses/video/112105171/112105171.html
5. https://fm-hfs-iitb.vlabs.ac.in/
   https://archive.nptel.ac.in/content/syllabus_pdf/105101082.pdf

CO	CO Description	PO	PSO
CO1	Explain fundamentals of fluid properties, statics, kinematics, dynamics, pipe flow, dimensional analysis, and hydraulic machines.	—	—
CO2	Apply fluid mechanics principles to compute forces, flow parameters, losses, and machine performance.	PO1 (2), PO2 (2)	PSO1 (2), PSO2 (2)
CO3	Analyze fluid systems involving statics, dynamics, pipe flow, and hydraulic machines to determine behavior.	PO2 (2), PO4 (2)	PSO1 (2), PSO2 (2)
CO4	Evaluate performance, efficiency, and operating conditions of fluid systems and machines.	PO3 (2), PO5 (2)	PSO1 (2), PSO2 (2), PSO3 (1)

(1) = Low; (2) = Medium; (3) = High

ME25401	Thermal Engineering – I	L	T	P	C
		3	0	0	3

Course Objectives:

The course focuses on applying thermodynamic laws to analyze thermodynamic and steam power cycles, while also providing an understanding of IC engines, their systems, and aircraft propulsion cycles.

Air Standard Cycles: Carnot, Otto, Diesel, Dual – Cycle Analysis, Comparison of Otto, Diesel and Dual Cycle, Atkinson cycle.

Activities: Write a program to analyze Otto, Diesel, and Dual cycles for given inputs (compression ratio, heat addition, etc.) and compute efficiency, work output, and mean effective pressure. Generate P–V and T–S diagrams for different air standard cycles and compare their performance.

Vapour Power Cycle: Carnot Cycle, Basic Rankine Cycle, Comparison, Rankine cycle Improvements - Reheat, Regenerative Cycle, Combined cycle power generation

Activities: Develop a program to analyze the Rankine cycle with and without reheat/regeneration, and compute efficiency, turbine work, and pump work. Plot T–S diagrams for basic and modified Rankine cycles and compare performance improvements.

Internal Combustion Engines: IC engine – Classification, working, components and their functions. Ideal and actual: Valve and port timing diagrams, P-V diagrams. Two stroke & four stroke, and SI & CI engines – comparisons. Combustion in SI & CI Engines – Abnormal combustion, Knocking and detonation.

Activities: Prepare a comparative performance report of Petrol and Diesel Vehicles.

Internal Combustion Engine Testing: Measurement of Performance parameters and characteristics, Performance calculations. Morse and Heat Balance tests

Activities: Develop a simple program or spreadsheet to estimate engine performance parameters (thermal efficiency, BSFC, etc.) for given input conditions.

Internal Combustion Engines Auxiliary System: Simple Carburettor, Fuel Injector, Multipoint Fuel Injection (MPFI), Gasoline Direct Injection (GDI) and Common Rail Direct lnjection (CRDI) systems, Electronic fuel Injection. Ignition system, Lubrication and Cooling systems. Concepts of Supercharging and Turbocharging.

Activities: Prepare a case study report on modern fuel injection systems (MPFI, GDI, CRDI) used in automobiles.

Gas Turbines Cycle: Gas turbine cycle analysis – open and closed cycle. Performance and its improvement - Regenerative, Intercooled, Reheated cycles and their combination

Activities: Write a program to analyze Brayton cycle and its modifications (intercooling, reheating, regeneration) and compute efficiency and work output. Plot T–S diagrams for different gas turbine cycles and compare their performance.

Jet Propulsion: Principle of Jet Propulsion, IC engine driven Propulsion, Gas turbine Propulsion, Turbojet, Turboprop, Turbofan engines.

Activities:

- Develop a basic computational model to calculate thrust, specific fuel consumption, and efficiency for turbojet or turbofan engines.
- Prepare a comparative analysis report on turbojet, turbofan, and turboprop engines based on real aircraft applications.

Weightage: Continuous Assessment: 40%, End Semester Examinations: 60%.

Assessment Methodology: Quiz - 10%, Assignments - 20%, Review of GATE/ESE Questions - 20% and Internal Examinations - 50%

References:

1. Mahesh M. Rathore, “Thermal Engineering”, 1st Edition (Latest Reprint), McGraw Hill Education, 2024.
2. Ganesan. V, “Internal Combustion Engines”, 4th Edition (Latest Reprint), McGraw Hill Education, 2024.
3. Ballaney. P. L, “Thermal Engineering”, 27th Edition, Khanna Publishers, 2024.
4. Rajput R.K., “Thermal Engineering”, 11th Edition, Laxmi Publications (P) Ltd., 2022.
5. Onkar Singh, “Applied Thermodynamics”, 5th Edition, New Age International Publishers, 2021.

E-resources:

1. https://onlinecourses.nptel.ac.in/noc25_me36/preview?utm_source=chatgpt.com
2. https://onlinecourses.nptel.ac.in/noc24_me03/preview?utm_source=chatgpt.com
3. https://online.vtu.ac.in/course-details/Applied-Thermodynamics?utm_source=chatgpt.com

CO	CO Description	PO	PSO
CO1	Explain the basic principles of thermodynamics, air standard cycles, vapour power cycles, IC engines, gas turbines and jet propulsion systems.	–	–
CO2	Apply thermodynamic laws to solve problems related to power cycles and performance of IC engines and gas turbines.	PO1 (2), PO2 (2)	PSO1 (2)
CO3	Analyse and compare the performance of air standard cycles, vapour power cycles, IC engines and gas turbine cycles including improvement methods.	PO2 (2), PO3 (2)	PSO2 (2)
CO4	Demonstrate testing and performance evaluation of IC engines and understand the working of engine auxiliary systems.	PO4 (2), PO5 (1)	PSO2 (2), PSO3 (1)

(1) = Low; (2) = Medium; (3) = High

ME25402	Manufacturing Processes – I	L	T	P	C
		3	0	2	4

Course Objectives:

Appreciate and determine the appropriate manufacturing pathways for producing specific products.

Digital storage of geometrical shapes: Geometrical shape types (2.5D – constant cross section, 2.75D – offsetting a surface, and general 3D shapes) Storage of shape information in digital geometrical models; Wireframe model, surface models and solid models. File formats; STL file; slicing into layers.

Practical:

1. Create Wire frame model, surface model and solid model of a given geometrical shape in any CAD software

Subtractive Manufacturing

Mechanics of Cutting: Orthogonal and Oblique cutting, Chip formation, Type of chips, Nomenclature of single point cutting tool, Shear plane theory, Shear stress, Shear strain, Forces in machining, Merchant's circle, cutting tool materials, Tool wear and mechanisms, cutting fluids, temperature in machining, surface roughness, tool life, machinability.

Turning, Drilling, Milling: Lathe Operations, Need for drilling, Geometry of twist drill, Types of drilling machines, Drilling and related operations, up milling and down milling, Types of Milling machines and cutters, milling operations, indexing, work holding and tool holding devices, machining time and power estimation

Abrasive Processes: Grinding – Specification of grinding wheel, mechanism of metal removal in grinding, grindability and its improvement, Dressing, truing, types of grinding machines– surface grinding, cylindrical grinding, centerless grinding, Honing, lapping, super finishing methods.

Practical:

1. Taper turning on a cylindrical workpiece using different methods (form tool method, compound rest method, and tailstock set-over method).
2. Evaluation of the machinability of two work materials based on tool wear and chip morphology and cutting force.
3. Evaluation of the grindability of two materials by determining the G-ratio and measuring the surface finish.
4. Drilling and Milling practicals
5. Use of any open source CAM software to simulate milling

Additive manufacturing: ISO nomenclature of AM processes; AM processes that add material in an empty space **-** directed energy deposition (DED), material extrusion (MEX), material jetting (MJT), AM processes that modify material already existing i - powder bed fusion (PBF), vat photopolymerization (VPP), Binder jetting (BJT)

Practical:

6. Simulation of 3D printing processes using any open source software

Programmable Positioning systems: Need for spatial positioning of cutting tools and Additive Manufacturing heads; tool paths as containing shape information; Coordinate systems, Motion control systems – point to point control, continuous path control, Coordinate systems, Interpolation methods, Absolute and incremental positioning, CNC software, DNC, Advantages of CNC.

Practical:

7. Manual part programming for CNC turning centre (facing, turning, threading).
8. Manual part programming for CNC machining centre (drilling, pocketing, contouring).

Pattern, Mold, and Mask making: Physical storage of surface information on molds, patterns and masks. Patterns and their types. Molds and their various features. Mold types: flow-through molds, open molds, closed molds; Making of masks for chemical machining, silicon electronics fabrication

Practical:

9. Make a 3D CAD model of any mold

Tasks:

T1. Design and manufacture a precision shaft component using lathe operations (turning, threading, knurling), ensuring dimensional accuracy and surface finish.

T2. Develop a machining process plan for producing a gear (spur/helical) using milling, shaping, or hobbing, including tool selection and machining parameters.

T3. Design and optimize a grinding process to achieve required surface finish and dimensional tolerance for a mechanical component (e.g., bearing shaft).

T4. Develop a CNC part program for a turning or machining center to manufacture a component (e.g., flange or bracket) using appropriate tool paths and cycles.

Weightage: Continuous Assessment: 50%, End Semester Examinations: 50%

Assessment Methodology:

Written Test (40%), Practical (30%), Activity (30%).

Practical: Lab Experiments (50%), Tasks (50%) (Each student must complete a minimum of two tasks).

Activity: Review of GATE/ESE Questions(10%), Mini Project/ Quiz/ Assignment Programs/ Flipped Class /Seminar Presentation (20%)

References:

1. Serope Kalpakjian and Steven Schmid, Manufacturing Engineering and Technology, 9th edition, Pearson, 2025
2. Groover, Mikell P. Fundamentals of modern manufacturing: materials, processes, and systems. 7th Edition, John Wiley & Sons, 2019.
3. Knight, Winston A., and Geoffrey Boothroyd. Fundamentals of metal machining and machine tools. CRC Press, 2019.
4. A. B. Chattopadhyay, Machining and Machine Tools, 2nd Edition, Wiley, 2017.
5. Juneja, B. L. Fundamentals of metal cutting and machine tools. New Age

International, 2017.
6. P N Rao, Manufacturing Technology: Metal cutting and machine tools. Tata McGraw-Hill, 2013.

E-resources:

1. https://nptel.ac.in/courses/112103245
2. https://nptel.ac.in/courses/112105233
3. https://nptel.ac.in/courses/112103250
4. https://nptel.ac.in/courses/112104290
5. https://fab-coep.vlabs.ac.in/exp/3d-machining/theory.html
6. http://vlabs.iitkgp.ac.in/vmp/exp1/index.html#
7. http://vlabs.iitkgp.ac.in/vmp/exp1/Webpage/index.html

	Description of CO	**PO**	**PSO**
CO1	Explain fundamental concepts of metal cutting, machine tools, abrasive processes, gear manufacturing, and CNC machines and programming.	–	–
CO2	Apply lathe, milling, drilling, shaping and broaching operations to manufacture precision components.	PO3 (2)	PSO1 (2)
CO3	Analyze metal cutting mechanics, tool geometry, and machining forces for practical applications.	PO2 (2)	PSO2 (1)
CO4	Assess the machinability of materials through qualitative observations and quantitative measurements.	PO4 (2)	PSO2 (2)

(1) = Low; (2) = Medium; (3) = High

ME25C10	Standards in Mechanical Engineering	L	T	P	C
		1	0	0	1

Course Objectives:

The course introduces standardization, conformity assessment, and the roles of BIS and ISO, while familiarizing students with key standards in engineering drawings, testing, management systems, and machinery safety for practical application.

Introduction: Aims of standardization, Types of standards, Importance of standards to industry.

Activity: Study and present the importance and types of standards used in mechanical engineering industries.

Conformity Assessment: Objects of conformity assessment, First-party assessment, Second-party assessment, Third-party assessment, Certifications and markings.

Activity: Analyze different types of conformity assessment and identify certification markings on products.

Bureau of Indian Standards (BIS): Objectives, roles, and functions of BIS, Application of any one BIS standard.

Activity: Study BIS functions and identify BIS standards applied to common engineering components.

International Organization for Standardization (ISO): Key principles for ISO standards development, ISO Directives, Stages in ISO standard development, Application of any one ISO standard.

Activity: Analyze ISO standard development process and study key ISO standards.

Standards for Engineering Drawing: SP 46:2003 – Sizes and layout of drawing sheets, Scales, Lines, Sections, Screw threads and threaded parts, Springs, Gears, Dimensioning, Surface texture, Linear and angular tolerances, Geometrical tolerances.

Activity: Interpret engineering drawings using SP 46 standards and identify symbols, tolerances, and dimensions.

ASTM Standards (American Society for Testing and Materials): Objectives and scope of ASTM standards, Application of any one ASTM standard.

ASME Standards (American Society of Mechanical Engineers): Objectives and scope of ASME standards.

Weightage: Continuous Assessment: 100%.

Assessment Methodology: Quiz - 10%, Assignments - 20%, and Internal Examinations - 70%

References:

1. Bureau of Indian Standards, Engineering Drawing Practice for Schools & Colleges, SP 46:2003, Bureau of Indian Standards, New Delhi, India, 2003, ISBN: 81-7061-019-2.
2. W. Hesser, A. Inklaar, and J. Weigand, Standardization Essentials: Principles and Practice, Wiley-VCH, Weinheim, Germany, 2010.
3. C. D. Sullivan, Standards and Standardization: Basic Principles and Applications, Marcel Dekker Archive Information, New York, USA, 1983, ISBN: 0-8247-1919-0.
4. ASTM International Standards Catalog, ASTM International, West Conshohocken, Pennsylvania, USA.
5. James D. McCabe, Introduction to the Practice of Mechanical Engineering, including ASME standards and codes, Pearson Education.

E-resources:

1. https://standardsbis.bsbedge.com/
2. https://www.iso.org/sites/edumaterials/trbsaunders.pdf
3. https://www.ds.dk/media/px5jhney/a-world-built-on-standards.pdf?srsltid=AfmBOoqD36gwxLHthlIwtvrIqVtSnLaQFL7N8vSJcNfrxSF_qUstL8ia
4. https://www.bis.gov.in/?lang=en
5. https://www.iso.org/home.html

CO	CO Description	PO	PSO
CO1	Explain fundamental concepts of standardization, conformity assessment, BIS, ISO, engineering drawing standards, testing standards, management systems, and machinery safety standards.	—	—
CO2	Explain the roles, functions, and applications of national and international standards in engineering practice.	PO1 (2)	PSO2 (2)
CO3	Apply principles of standardization and conformity assessment to identify and use appropriate standards, codes, and certifications in engineering applications.	PO1 (2), PO2 (2)	PSO1 (2), PSO2 (2)

(1) = Low; (2) = Medium; (3) = High

EN25C04	English Communication Skills Laboratory– II	L	T	P	C
		0	0	2	1

Course Objectives: The objectives of the course are to build students' advanced communication skills for workplace readiness and develop intercultural competence for effective collaboration in global and virtual teams. Prepare students for competitive exams with focused skill-building and test-oriented practice.

List of Activities

Stage Ready – Impactful Public Speaking

(i) Simulate a formal event such as an academic conference, convocation, or awards ceremony, where students roles including Master of Ceremonies (MC), Role as a dignitary, and a Commentator
(ii) Visual Prompt Storytelling: Use random images to create spontaneous stories, focusing on plot, setting, and character,
(iii) **Digital Presentation - Record** a short video explaining a project or technical concept, using slides, voiceover, and visual aids (to be uploaded using google classroom or drive link)

Professional and Application-Oriented Writing

(i) Résumé Preparation: Design ATS-friendly résumés tailored to various job descriptions, using action verbs and quantifiable impact. ·
(ii) Design engaging content for poster presentation relevant to their domain.

Receptive Skills in Workplace Communication·

(i) Reading articles related to their domain and discuss in groups
(ii) Visit company websites, make inferences and present in the class
(iii) Listen to recorded mock interviews and take detailed notes. Summarise key points and action items in a professional format and make a presentation.

Intercultural Communication

(i) Assertive vs Aggressive communication
(ii) Role play activities – workplace communication in intercultural/crosscultural contexts

From Campus to Career: Industry Skills and Global Exam Preparation

(i) Participate in HR interviews using AI tools or peer interviewers, responding to behavioural questions using methods like STAR (Situation, Task, Action, Result)
(ii) Practice Verbal Ability in competitive exams like UPSC, SSC, CDS, TNPSC, etc.

Weightage: Continuous Assessment: 60%, End Semester Lab Examinations: 40%.

Internal Assessment Methodology: 1. Oral story telling using visual prompts (30 marks)
2. Poster presentation (40 marks)
3. ATS resume writing (30 marks)

End Semester Assessment:

1. Interview (50 marks)
2. Verbal Ability test (50 marks)
   (students must bring the resume but evaluation must be done based on the performance in the interview)

References:

1. Lucas, Stephen, and Paul Stob. The Art of Public Speaking. Thirteenth edition, McGraw- Hill Education, 2020.
2. Abrahams, Matt. Think Faster, Talk Smarter: How to Speak Successfully When You're Put on the Spot. Simon & Schuster, 2023.
3. Beshara, Tony. Powerful Phrases for Successful Interviews, Rev. ed., McGraw-Hill, 2023.
4. Papalia, Anna. Interviewology: The New Science of Interviewing. Harper Business, 2024.
5. Verbal Ability and Reading Comprehension by Ajay Singh McGraw Hill Education 2020

E-resources:

1. Purdue OWL – Online Writing Lab (Academic and professional writing help) https://owl.purdue.edu/
2. Canva Resume Builder (Creative, ATS-friendly resume design) https://www.canva.com/resumes/
3. BBC Learning English – Pronunciation https://www.bbc.co.uk/learningenglish/english/features/pronunciation
4. India Bix website

	Description of CO	PO	PSO1
CO1	Understand basic industry-related reading materials.	-	-
CO2	Design and present a domain specific poster	PO9(3)	PSO1(2) PSO3(3)
CO3	Deliver effective digital presentations	PO9(3)	PSO2(1)
CO4	Communicate appropriately in intercultural/cross cultural contexts	PO9(3) & PO11(1)	PSO3(3)
CO5	Perform in interviews and competitive exams successfully	PO9(3)	PSO3(1)

(1) = Low; (2) = Medium; (3) = High